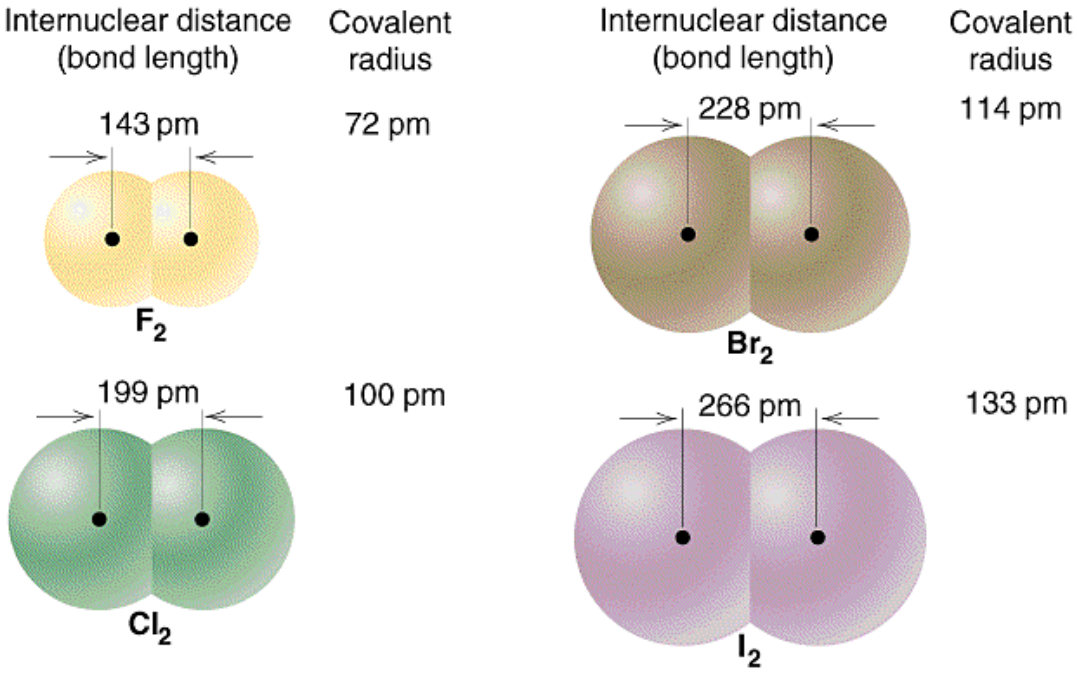
Lecture 9
Chemical Bond (Advanced)

第九讲 化学键

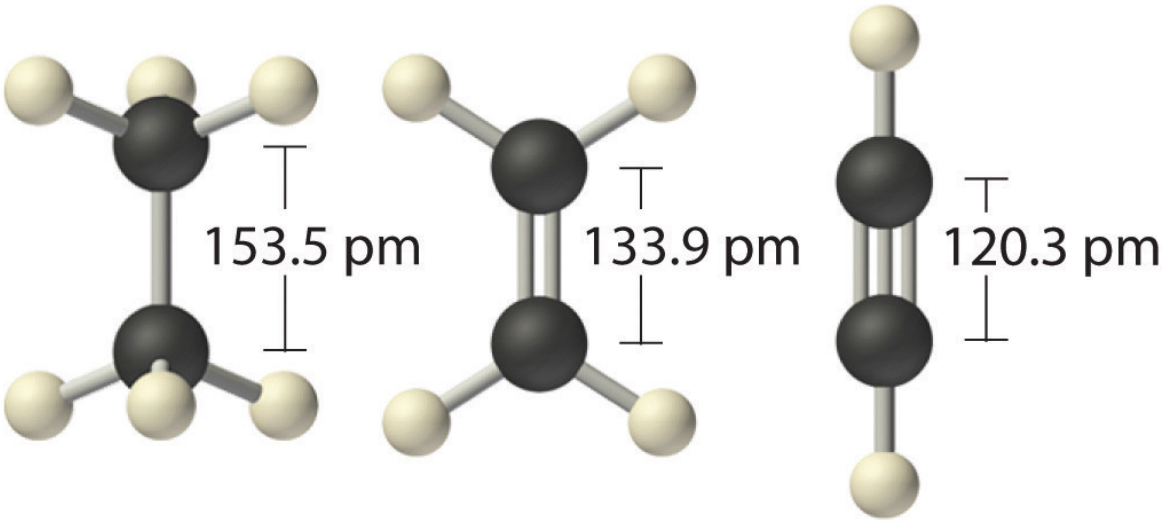
Bond Order, Length and Energy

Bond order describes a single, (bond order 1) double (BO 2) or triple (BO 3) bond. The higher the bond order the greater the strength of the bond and the shorter its length (for similar atoms).

The length of the covalent bond between two atoms can be approximated as the sum of the covalent radii of the two atoms.

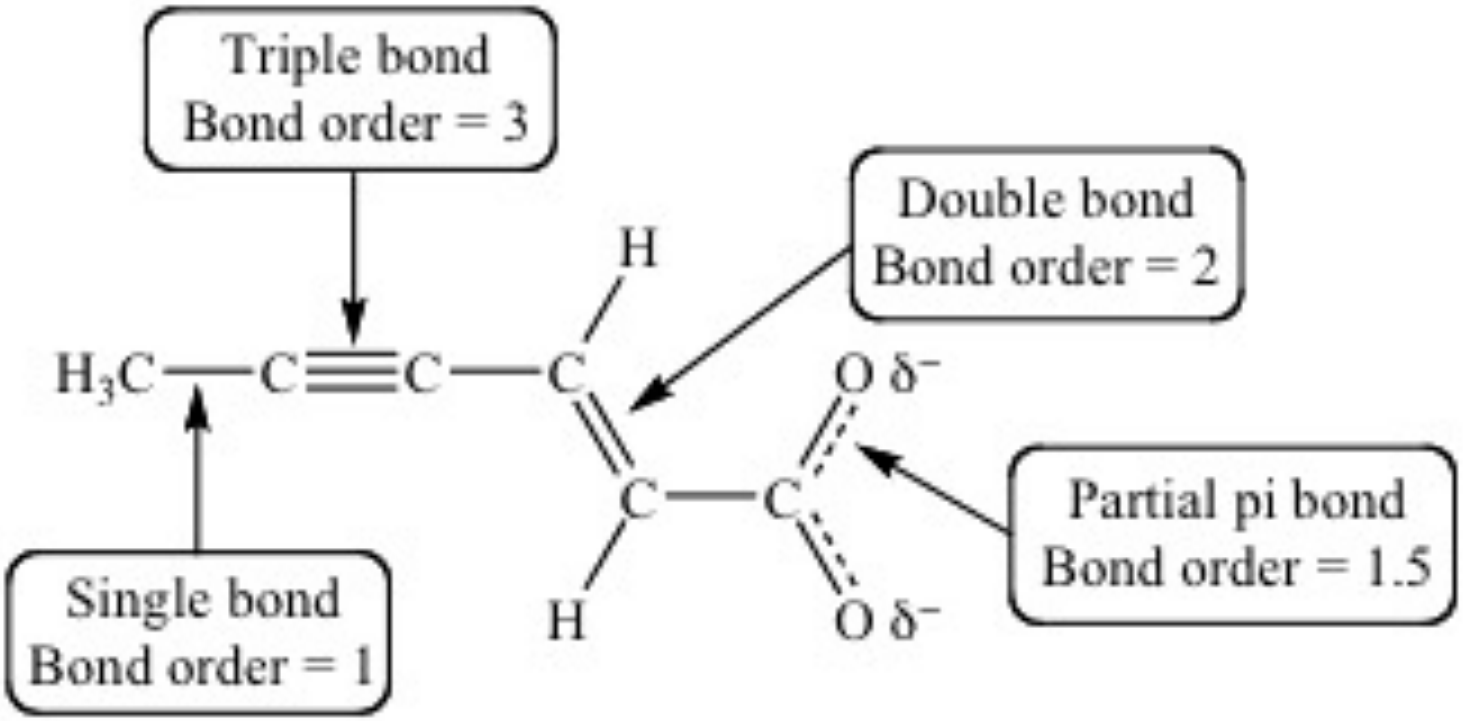


Bond	Bond Length (pm)	Bond	Bond Length (pm)
H—H	74	C—C	154
Cl—Cl	199	C = C	134
H—Cl	127	C ≡ C	120
C—H	109	O = O	120
		O—H	96



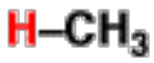
Single bond > Double bond > Triple bond

Bond order describes if the bond is single (bond order = 1), double (bond order = 2) or triple (bond order = 3)



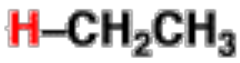
Bond Dissociation Energy
键的解离能

Bond dissociation energy is the amount of energy required to break 1 mole of covalent bonds in a gaseous species.



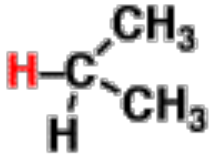
104 kcal/mol

Methane



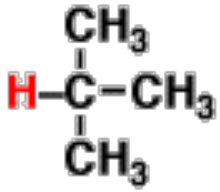
98 kcal/mol

Ethane



96 kcal/mol

Propane



93 kcal/mol

Butane

Average bond dissociation energies

Average Bond Enthalpies (kJ/mol)

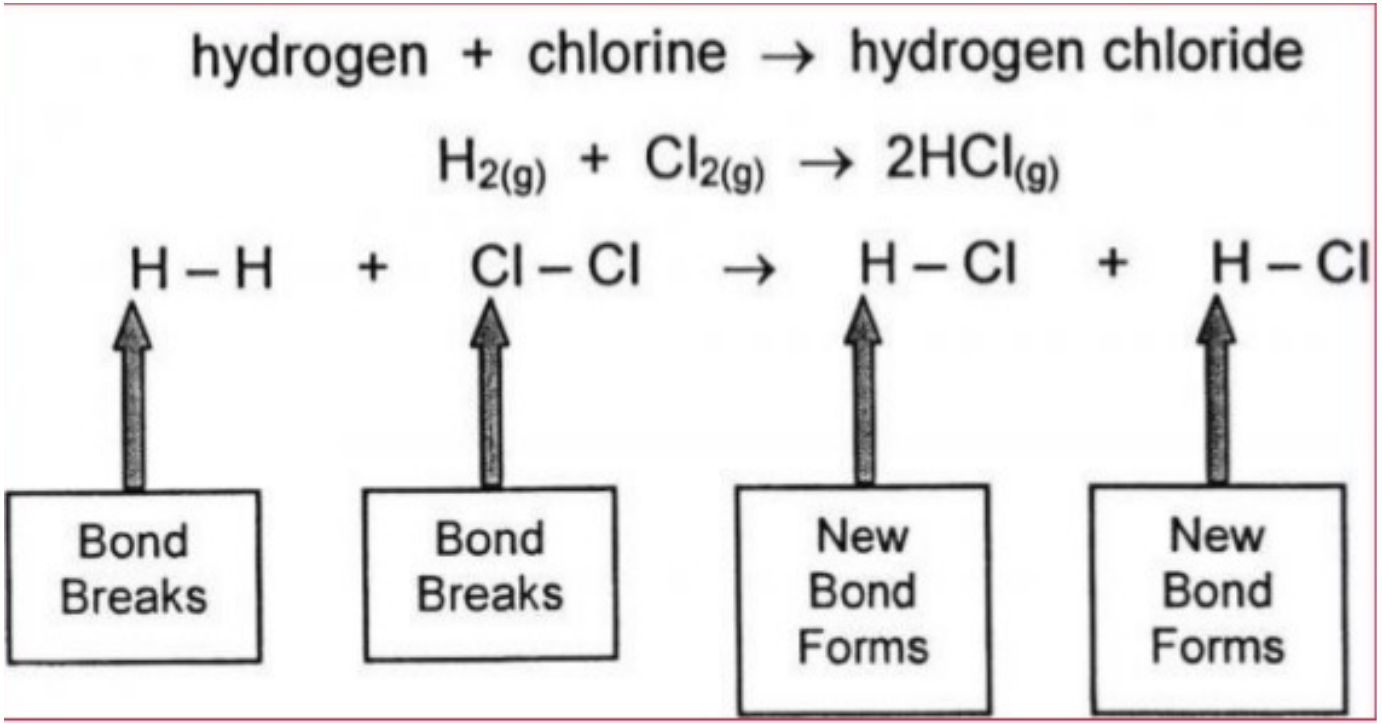
Single Bonds

C—H	413	N—H	391	O—H	463	F—F	155
C—C	348	N—N	163	O—O	146		
C—N	293	N—O	201	O—F	190	Cl—F	253
C—O	358	N—F	272	O—Cl	203	Cl—Cl	242
C—F	485	N—Cl	200	O—I	234		
C—Cl	328	N—Br	243			Br—F	237
C—Br	276			S—H	339	Br—Cl	218
C—I	240	H—H	436	S—F	327	Br—Br	193
C—S	259	H—F	567	S—Cl	253		
		H—Cl	431	S—Br	218	I—Cl	208
Si—H	323	H—Br	366	S—S	266	I—Br	175
Si—Si	226	H—I	299			I—I	151
Si—C	301						
Si—O	368						

Multiple Bonds

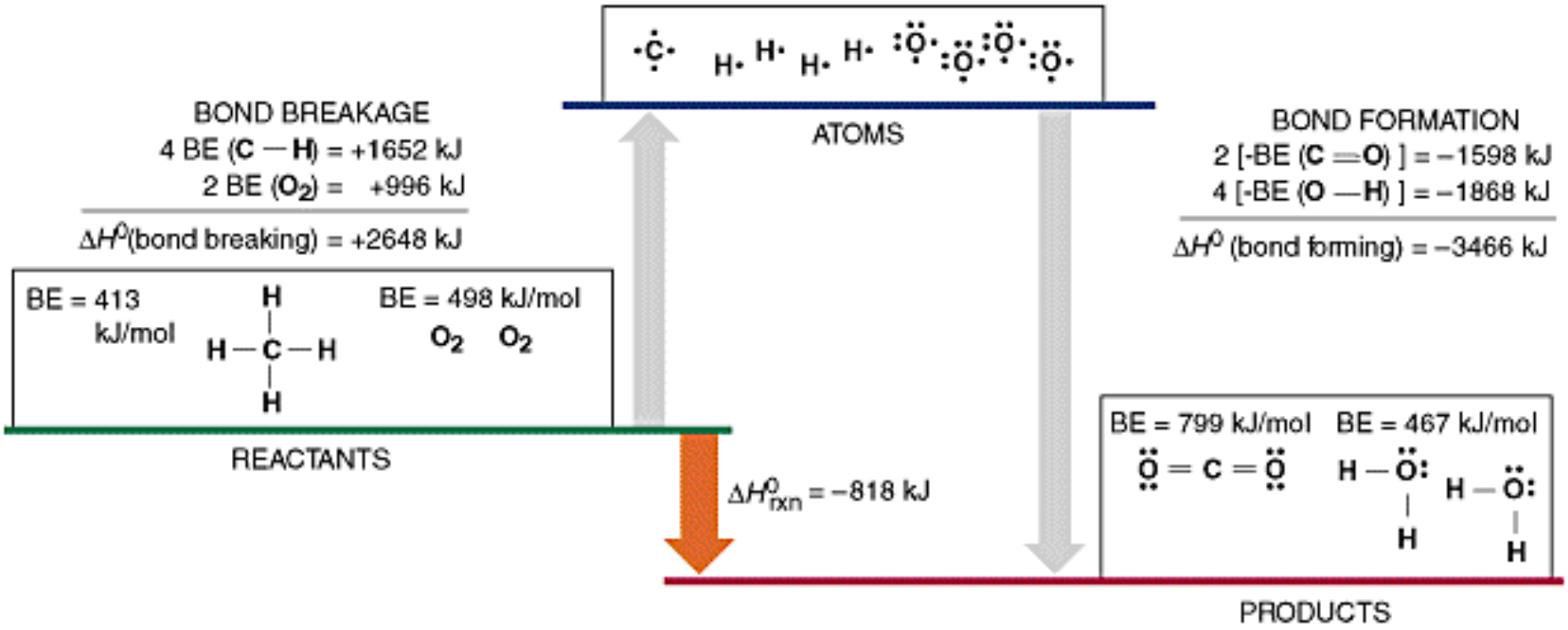
C=C	614	N=N	418	O_2	495
C≡C	839	N≡N	941		
C=N	615			S=O	523
C≡N	891			S=S	418
C=O	799				
C≡O	1072				

A chemical reaction is the breaking and making of bonds



Enthalpy of reaction from bond energies

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$



The reaction is exothermic

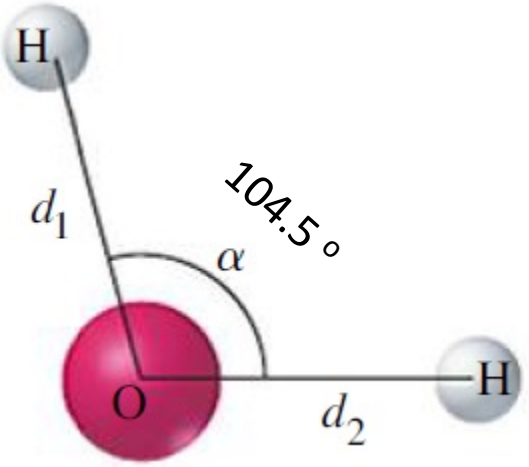
Molecular geometry



Determining the shapes of molecules



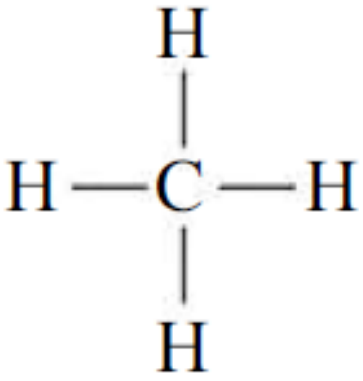
vs.



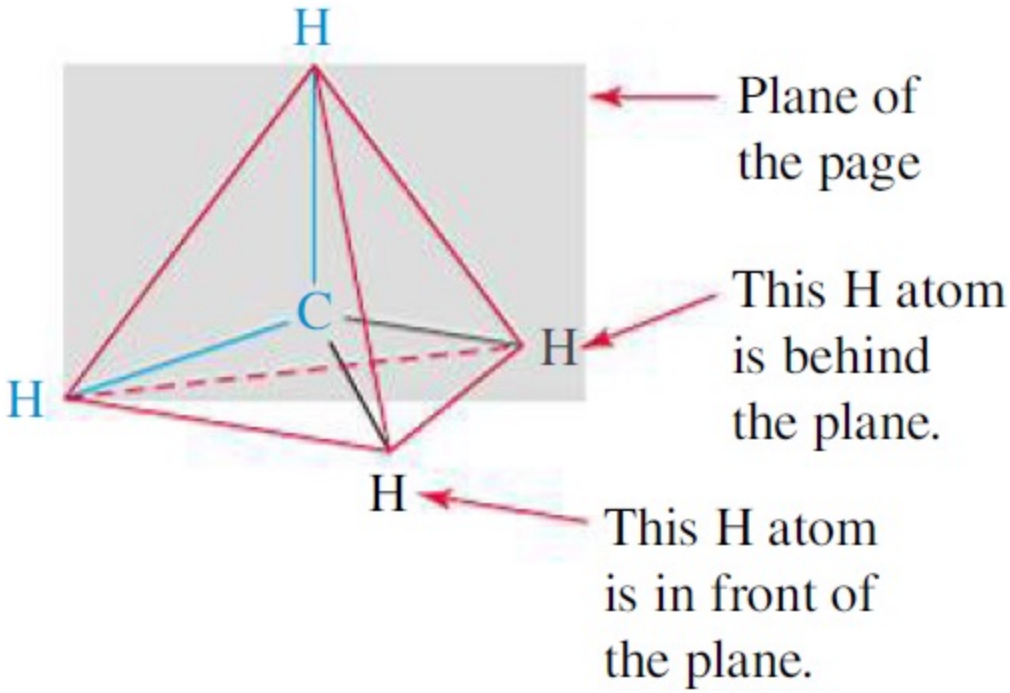
The fact that H_2O is bent accounts for many of the characteristics of water (e.g. high boiling point, high surface tension)

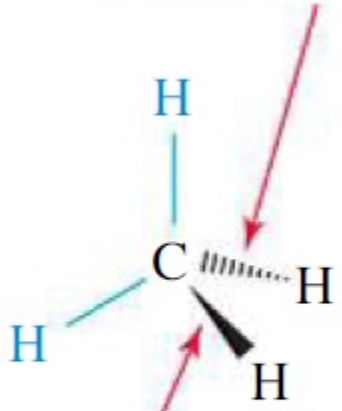
- Diatomic molecules are linear by definition.
- Triatomic molecules (such as H_2O) have two bonds and one bond angle. If the angle is 180° the molecule is linear. Any other angle makes the molecule “bent”, “angular” or “V-shaped”.
- Poly-atomic molecules (>3 atoms) have typically a three-dimensional geometric shape.

Methane (CH_4)



Tetrahedral shape

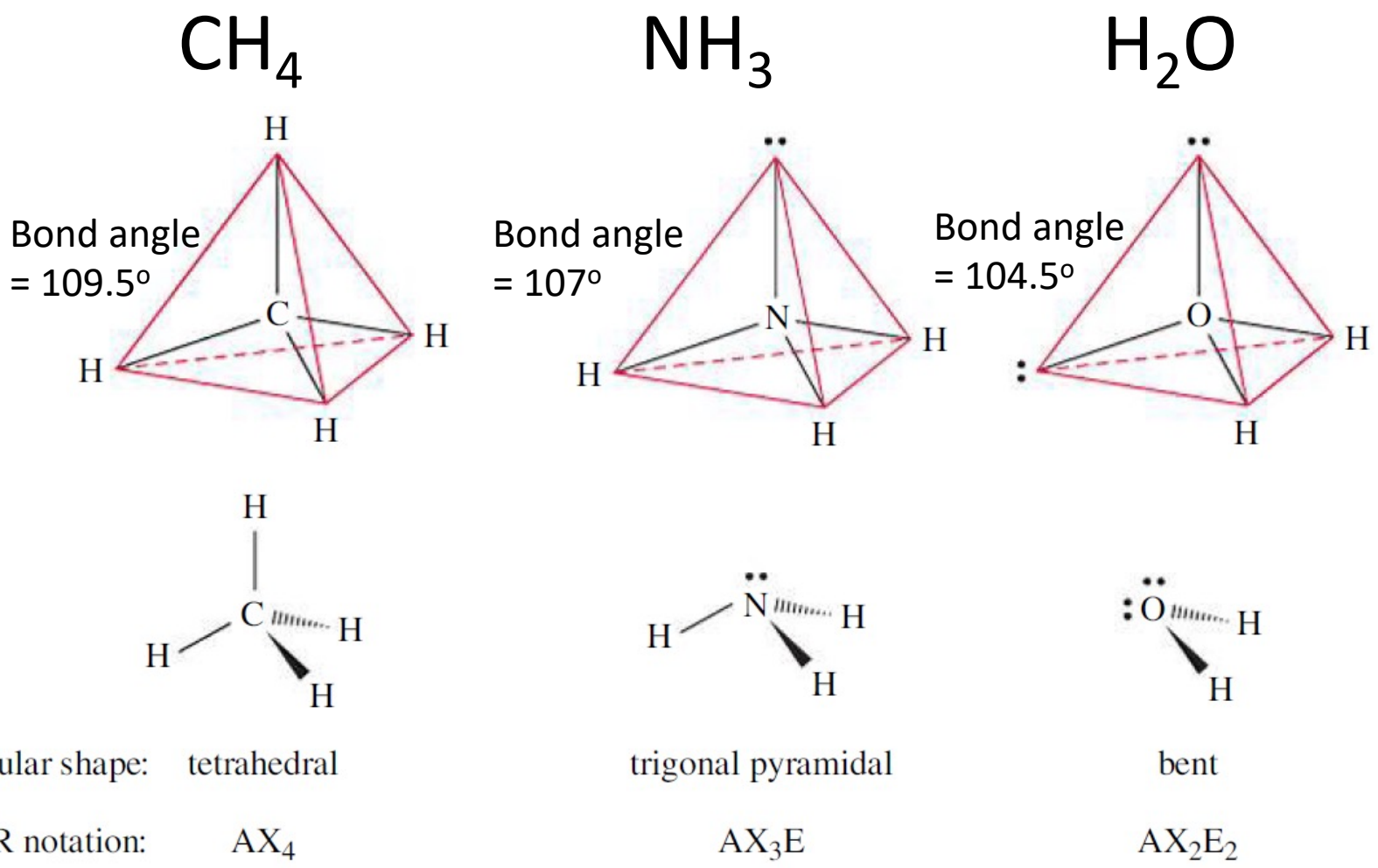




The dashed wedge represents a bond that points behind the plane of the page.

The solid wedge represents a bond that points out of the plane of the page.

Molecular shapes based on tetrahedral "electron-group" geometry



Valence shell electron pair repulsion theory

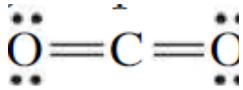
VSEPR (valence shell electron-pair repulsion) theory (价层电子对互斥理论)

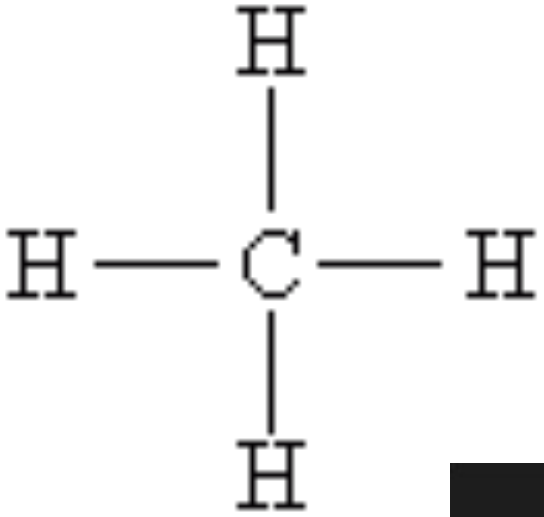
Assists in approximating the 3-D shape of molecules

The theory centers round **pairs of electrons and "electron groups"** in the valence electron shell of the central atom in the structure.

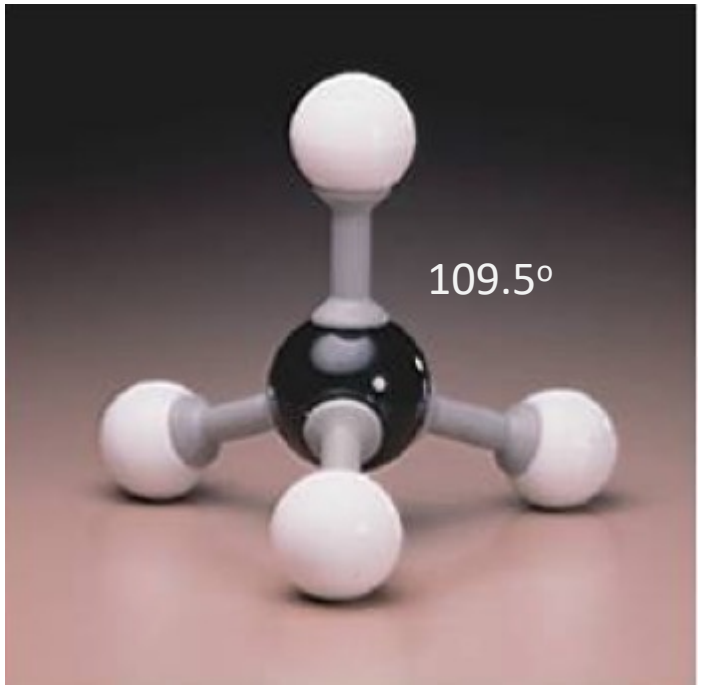
Electron pairs (whether in a bond or as lone pairs) repel each other and thus assume geometrical orientations to minimize the overall repulsion (they arrange themselves to be as far apart as possible).

Electron groups may be a pair (lone or bonded) but also a single unpaired electron or an atom with an incomplete octet. A group can also be a double or triple bond between two atoms.

For example, in  the central C atom is surrounded by two electron groups, each consisting of a double bond.



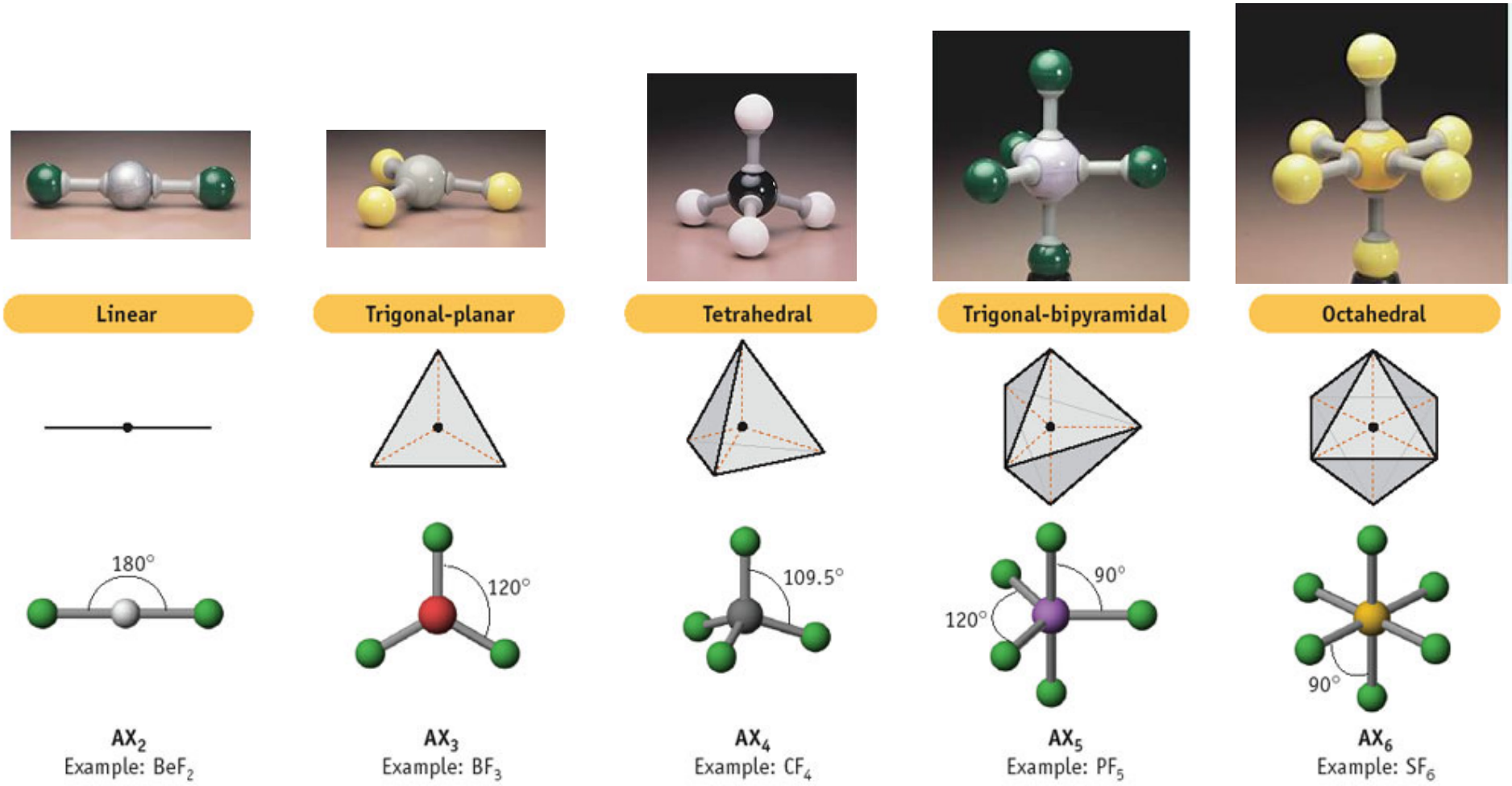
C in Methane is surrounded by 4 electron groups, hence its geometry is of a tetrahedral



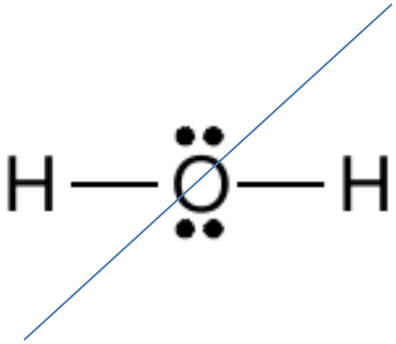
https://winter.group.shef.ac.uk/vsepr/CH4.html

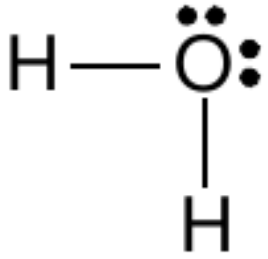
Valence Shell Electron Pair Repulsion (VSEPR)

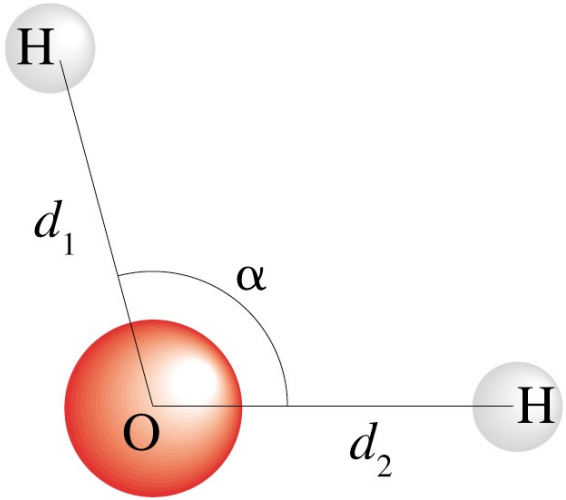
Electron pairs (in bonds or as lone pairs) repel each other and assume orientations about a central atom in order to minimize repulsions

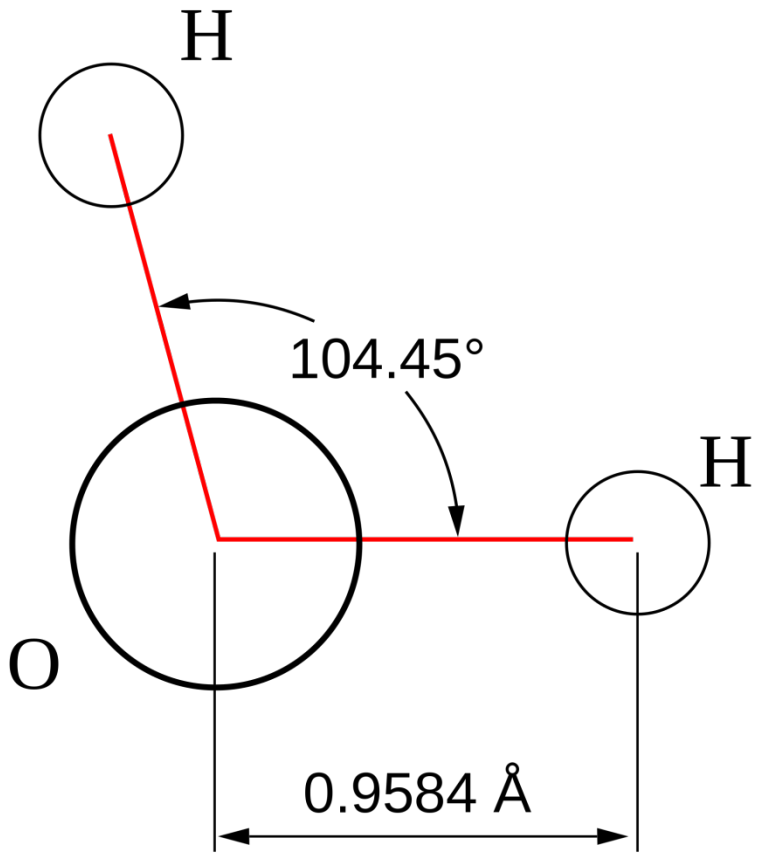


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H
O
H


H
O
H

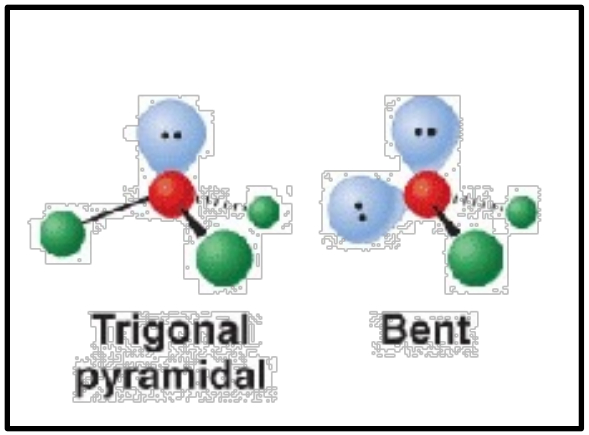

H
d_1
α
O
H
d_2

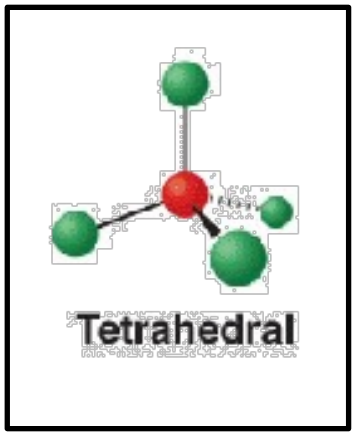

H
104.45°
H
O
0.9584 Å

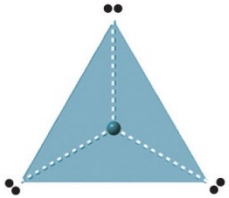
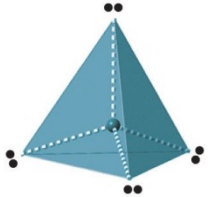
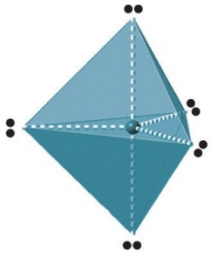
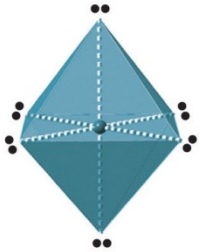
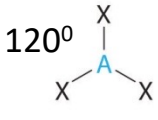
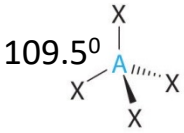
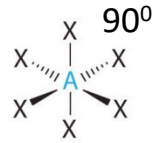
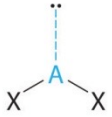
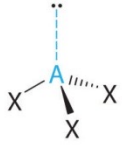
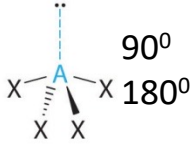
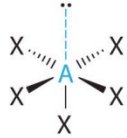
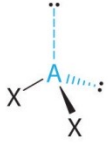
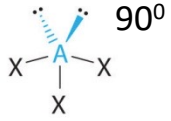
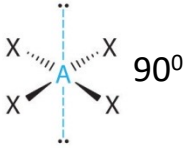
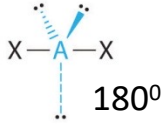
Geometry

- **Electron-group geometry** is the overall geometry including all lone pairs

- **Molecular geometry** is the geometry of the bonded atoms, ignoring lone pairs



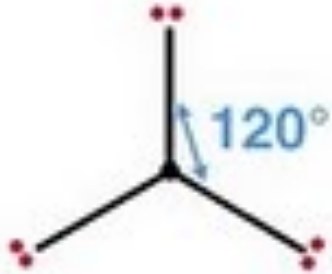


Electron Groups	2	3	4	5	6
Molecular Geometry					
	Linear	Trigonal planar	Tetrahedral	Trigonal bipyramidal	Octahedral
Zero Lone Pairs					
	Linear AX_2	Trigonal planar AX_3	Tetrahedral AX_4	Trigonal bipyramidal AX_5	Octahedral AX_6
One Lone Pair					
		Bent (V-shaped) AX_2E	Trigonal pyramidal AX_3E	Seesaw AX_4E One axial lone pair	Square pyramidal AX_5E
Two Lone Pairs					
			Bent (V-shaped) AX_2E_2	T-shaped AX_3E_2 Two axial lone pairs	Square planar AX_4E_2
Three Lone Pairs					
				Linear AX_2E_3 Three axial lone pairs	

Ideal Bond Angle



Linear



Trigonal planar



Tetrahedral

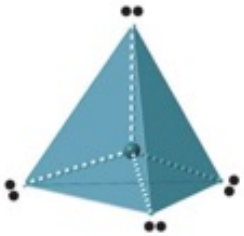


Trigonal bipyramidal



Octahedral

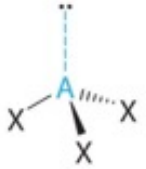
Deviations from ideal bond angle



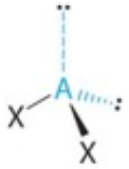
Tetrahedral



Tetrahedral AX_4

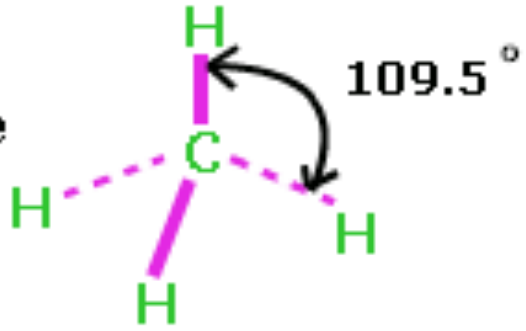


Trigonal pyramidal AX_3E



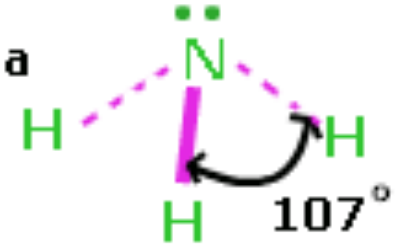
Bent (V-shaped) AX_2E_2

Methane



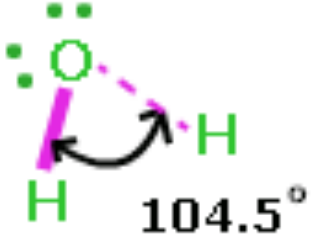
No lone pairs

Ammonia

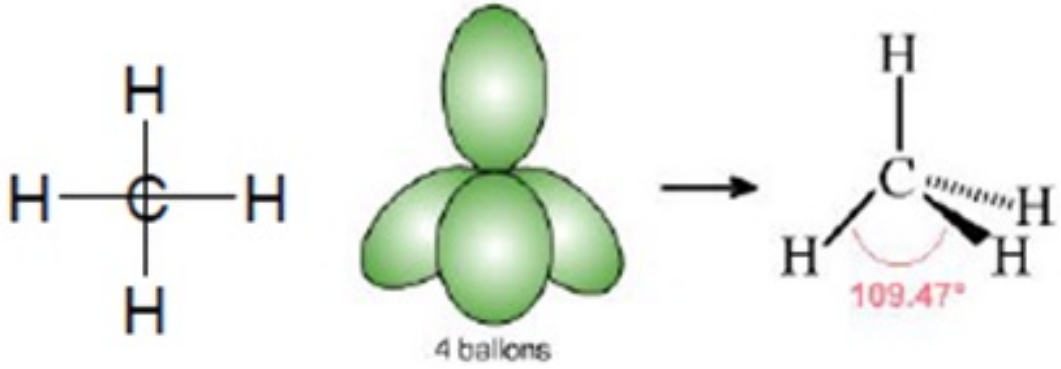


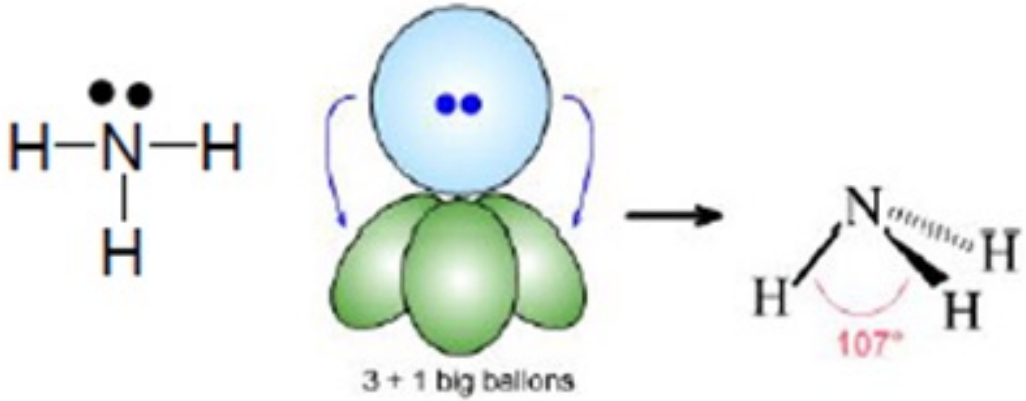
One lone pair

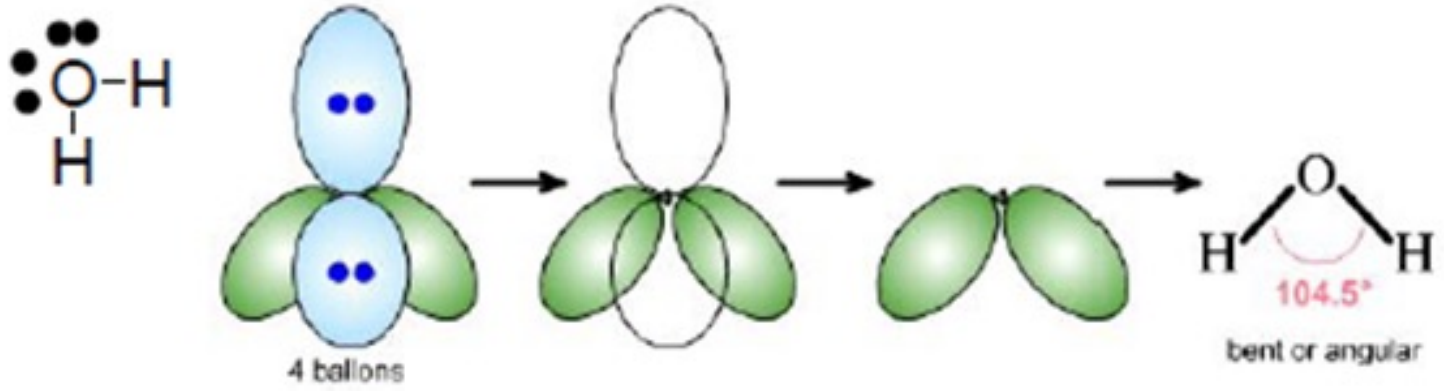
Water



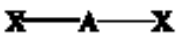
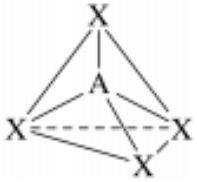
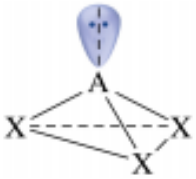
2 lone pairs

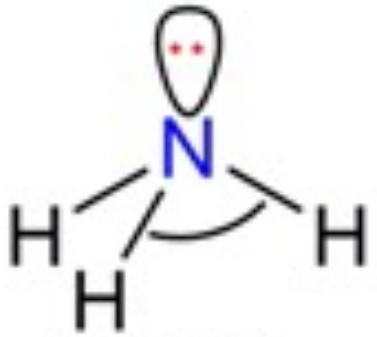

H
H
C
H
H
4 ballons
H
C
H
H
H
109.47°


H
N
H
H
3 + 1 big ballons
N
H
H
H
107°

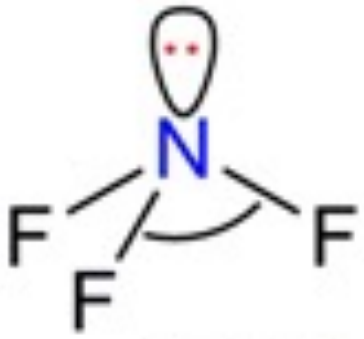

O
H
H
4 ballons
O
H
H
104.5°
bent or angular

Geometry plus approximated angles between the atoms

# of Groups of Electrons	*Electron Group Geometry*	*Number of Lone Pairs*	*Molecular Geometry*	*Approximate Bond Angles*	*Example Compound*
2	Linear	0	Linear 	180°	carbon dioxide, CO_2
3	Trigonal Planar	0	Trigonal Planar 	120°	formaldehyde, CH_2O
4	Tetrahedral	0	Tetrahedral 	109.5°	methane, CH_4
4	Tetrahedral	1	Trigonal Pyramid 	107°	ammonia, NH_3
4	Tetrahedral	2	Angular (Bent) 	104.5°	water, H_2O



107.3°



102°



104.5°



103.2°

Steps in predicting a geometric shape (based on VSEPR theory)

1. Draw a Lewis structure of the species
2. Determine number of electron groups around the central atom (distinguish between bond pairs and lone pairs)
3. Define the electron group geometry (linear, trigonal planar, tetrahedral, trigonal bi-pyramidal, octahedral) based on the rules on slide 18.
4. Determine molecular geometry from the positions of the atoms bonded directly to the central atom based on the rules on slide 18.

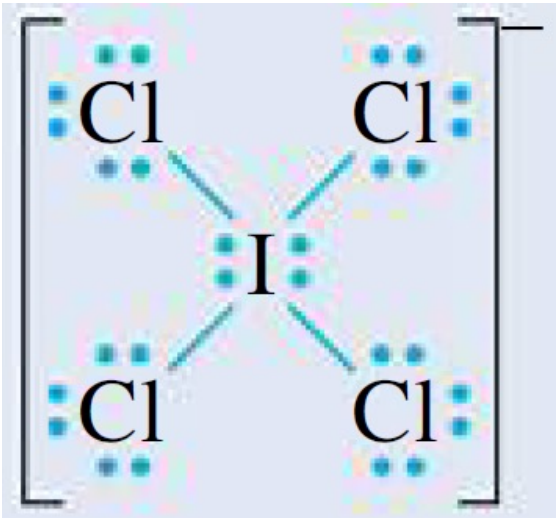
Example: predict the shape of ICl_4^-

1. Lewis structure: Valence electrons = 7 (I) + 4*7 = 28 (4 Cl) + 1 (molecule charge) = 36

- I is less electronegative than Cl, thus it is placed as the central atom.

- To account for all the 36 valence electrons we need to place additional 4 electrons around the I atom, as lone pairs (expands the valence shell to 12 e^-).

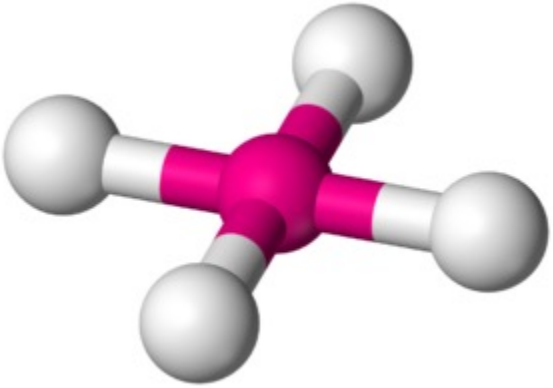
Recall variations to the octet rule



Step 2. There are six electron groups around the central (iodine) atom, four bond pairs and 2 lone pairs.

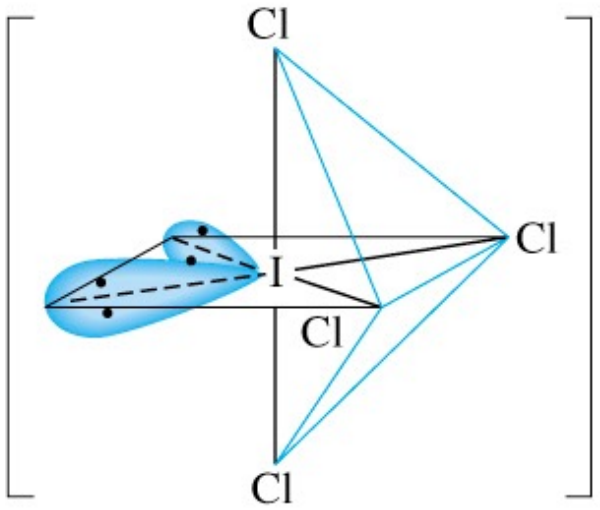
Step 3. Based on the Table on slide 18 the electron group geometry is octahedral.

Step 4. Based on the Table on slide 18, the ICl_4^- molecular geometry is square planar.

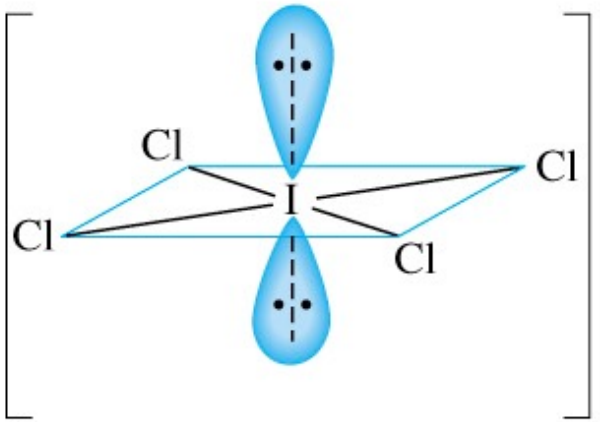


Idealized presentation

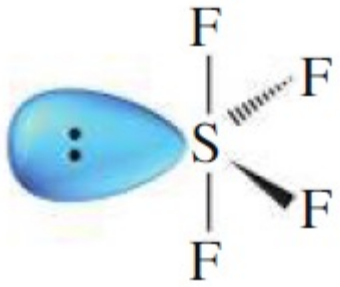
The structure always arranges itself to minimize repulsions (lone pairs dominate!)



(Incorrect)

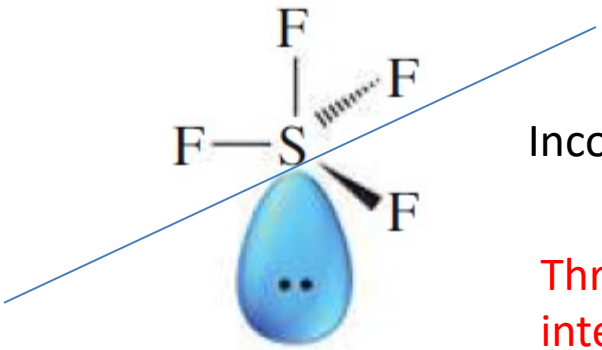


(Correct)



Correct

Two lone pair interactions at 90°



Incorrect

Three lone pair interactions at 90°

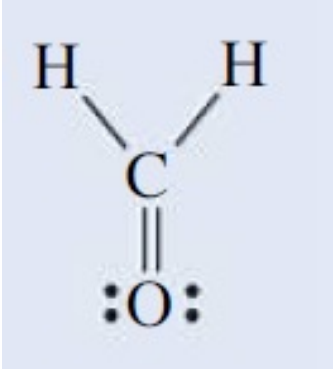
Order of repulsive forces (strongest to weakest):

Lone pair – lone pair > lone pair – bond pair > bond pair – bond pair

Predicting the shape of a molecule with a double or triple bond

Predict the geometry of formaldehyde (H_2CO)

Lewis structure:

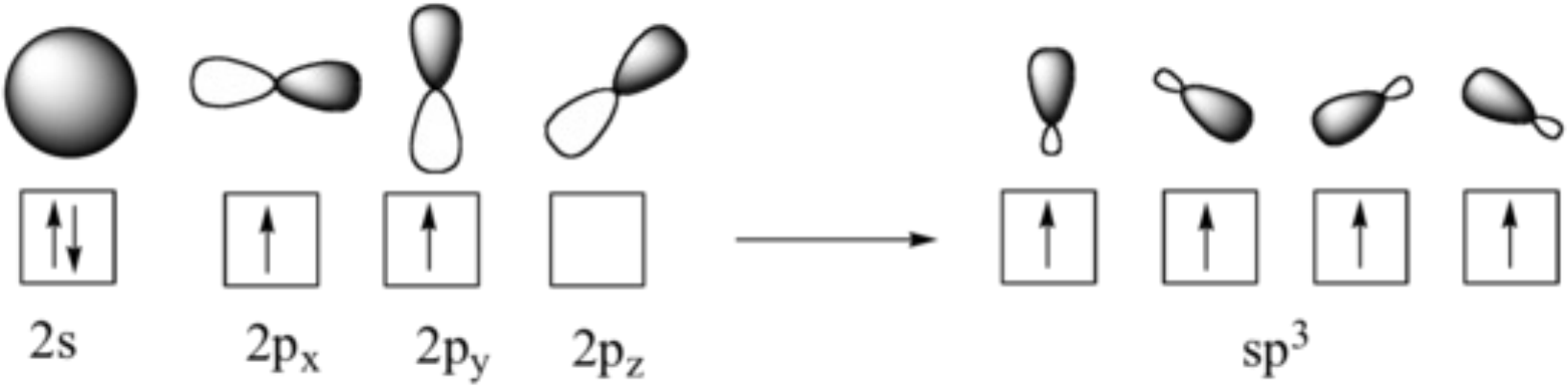


Electron groups: Double or triple bonds are treated as a single electron bond. There are thus 3 electron groups around the C atom.

From the VSEPR Table (slide 18) (3 e^- groups, no lone pairs) → trigonal planar (AX_3). Molecular angle = 120°.

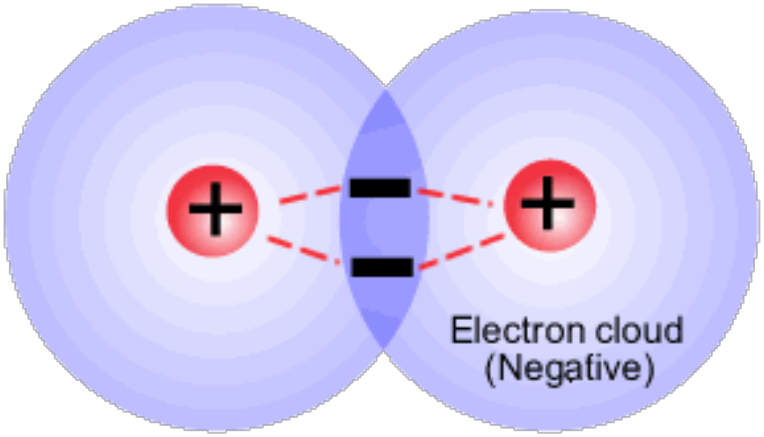
From this slide, for advanced chemical bonding understanding only!

Hybridization (杂化)

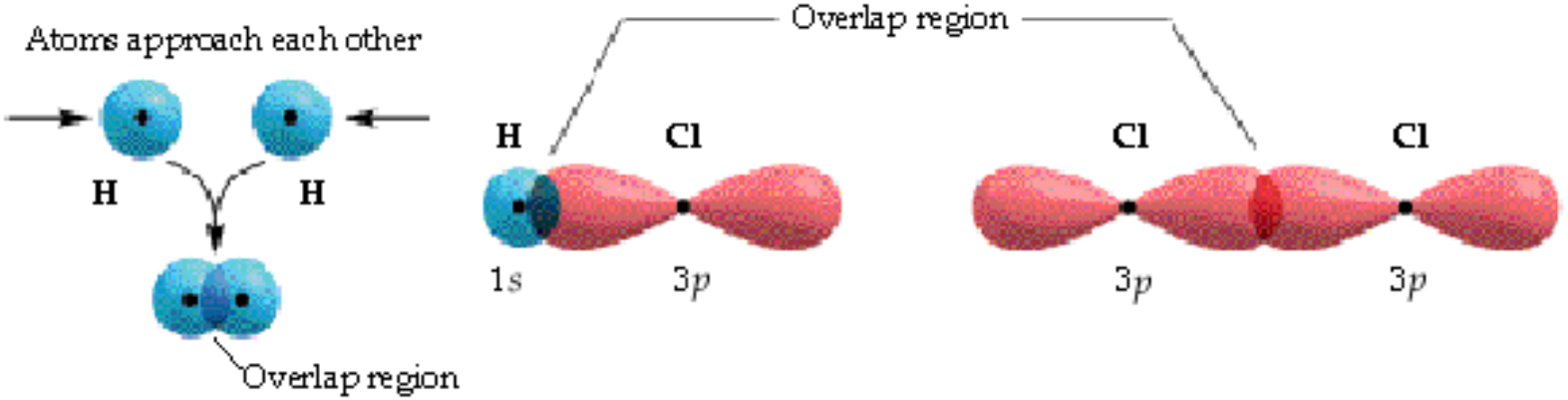


The description of covalent bond formation in terms of atomic orbital overlap is called the valence-bond method

The electrons experience a force of attraction from both nuclei. This negative - positive - negative attraction holds the two particles together



This attraction is called a chemical bond
one pair of electrons constitutes ONE bond



Hybridization of atomic orbitals

The valence-bond method was found not to represent reality in complex molecules!

Hybridization theory

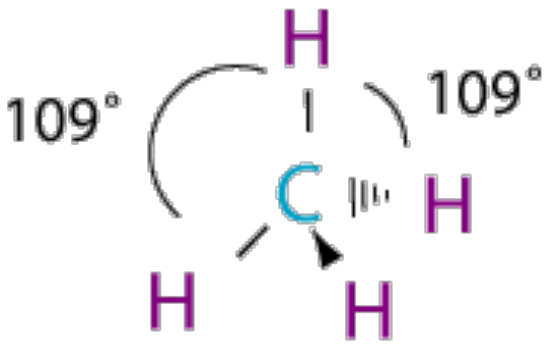
Atomic orbitals are mathematical expressions of the electron waves in atoms. An algebraic combination of the wave functions of the 2s and the three 2p orbitals in the carbon atom (for example) produces a new set of four identical orbitals, directed in a tetrahedral fashion.

The new orbitals (hybrid orbitals termed sp^3) have energies that are intermediate between the 2s and 2p orbitals (25% s character and 75% p).

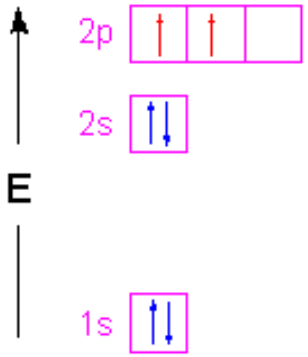
Important points

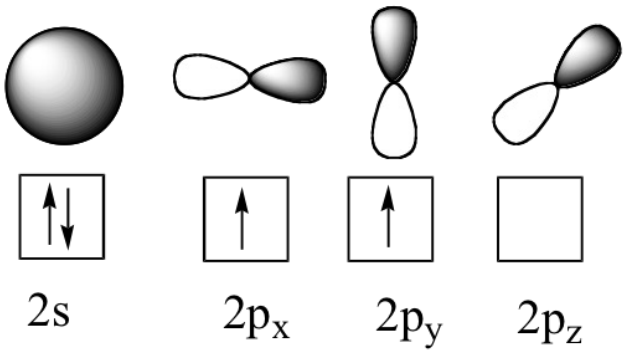
- Hybridization is a mathematical technique to rationalize experimentally observed shape of molecules (particularly carbon including molecules).
- It is not an actual physical phenomenon (we cannot observe electron charge distributions changing from those of pure orbitals to those of hybrid orbitals).
- For some covalent bonds no hybridization scheme works well.
- However, the concept works well for carbon containing molecules and therefore used extensively in organic chemistry.

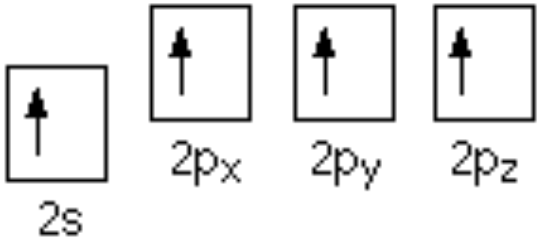
sp^3 hybridization of carbon atom in CH_4



Tetrahedral

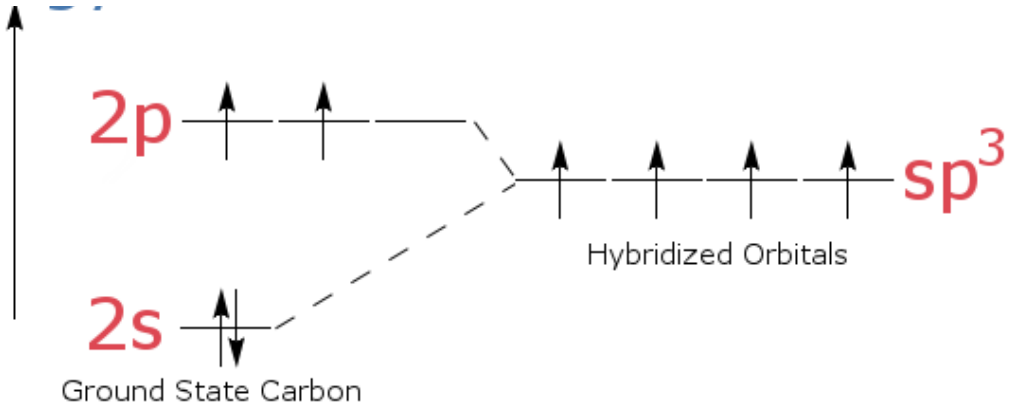


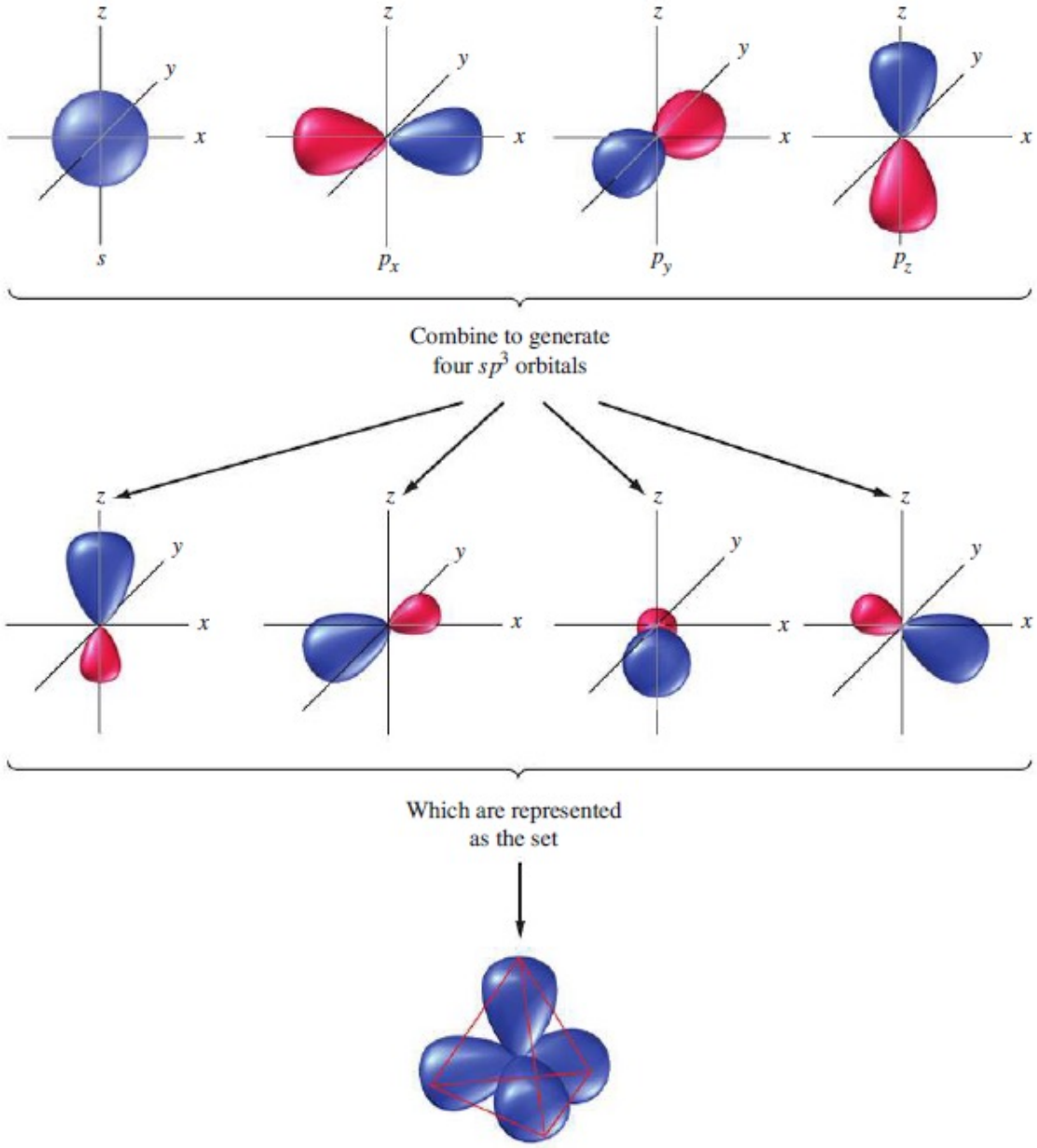




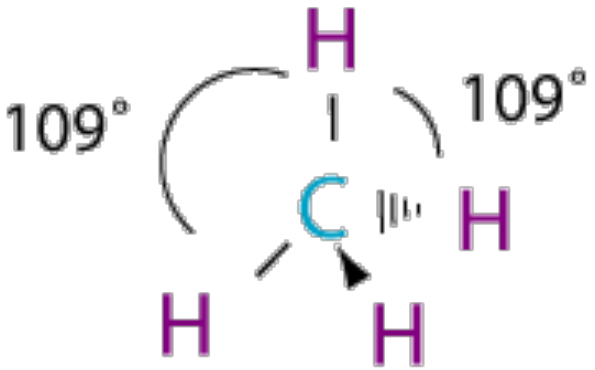
(Excited state)

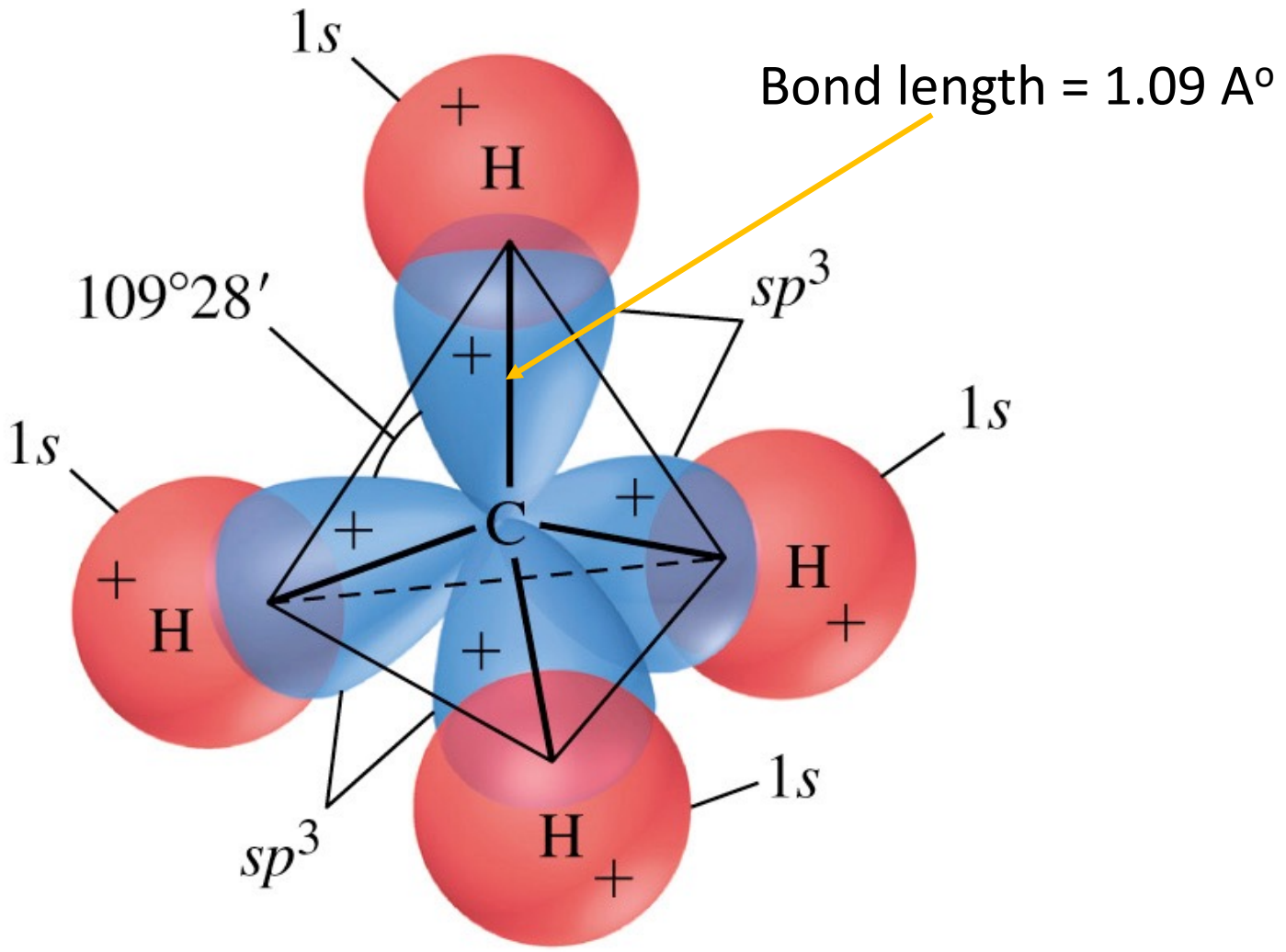
Hybridized representation





CH_4 is a stable molecule with 4 similar bonds



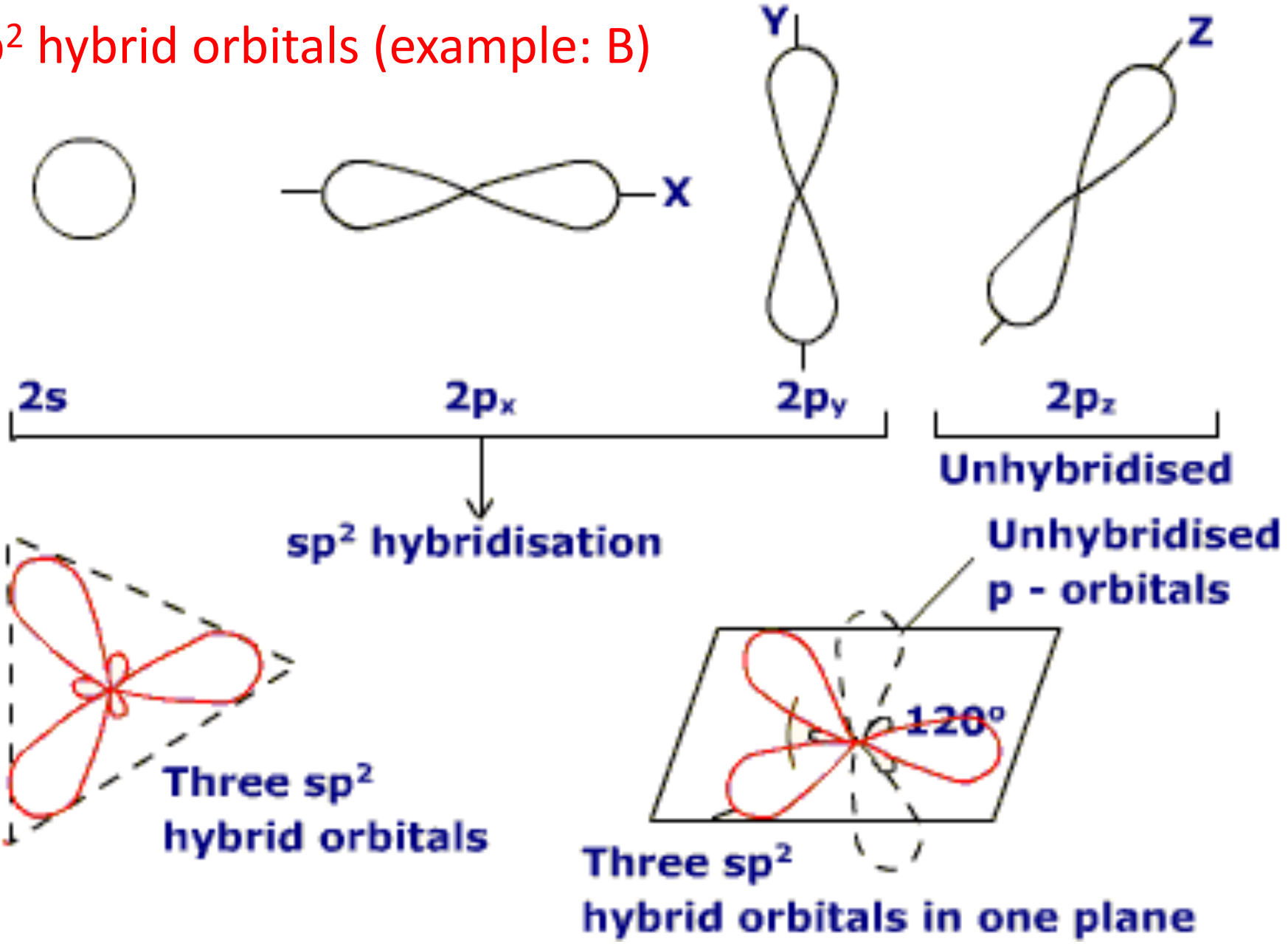

Bond length = 1.09 A°
1s
H
109°28′
sp3
1s
1s
H
C
H
H
sp3
1s
H

sp^2 hybridization

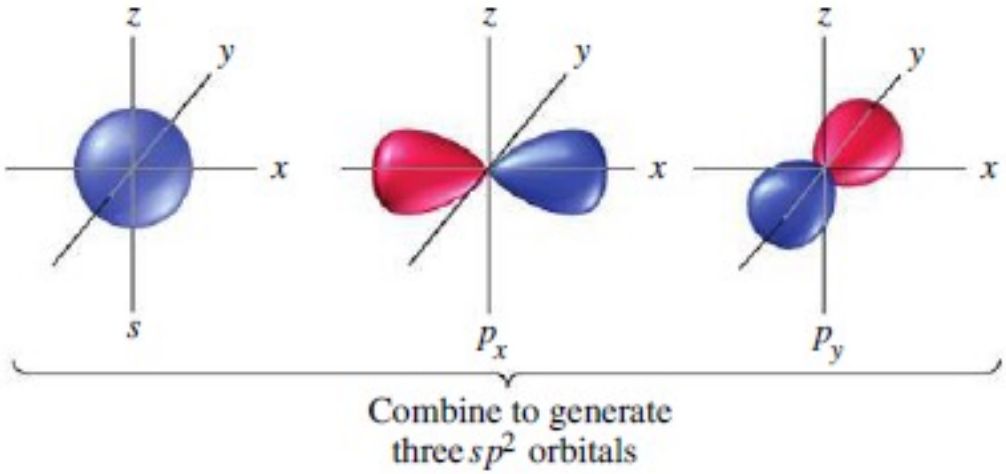
B has 4 orbitals ($2s^2 2p^1$) but only 3 valence electrons. For most boron compounds it is appropriate to combine the 2s and two 2p orbitals to get three sp^2 orbitals and leave one p orbital un-hybridized.

The sp2 configuration corresponds to trigonal – planar electron group geometry and 120° bond angles.

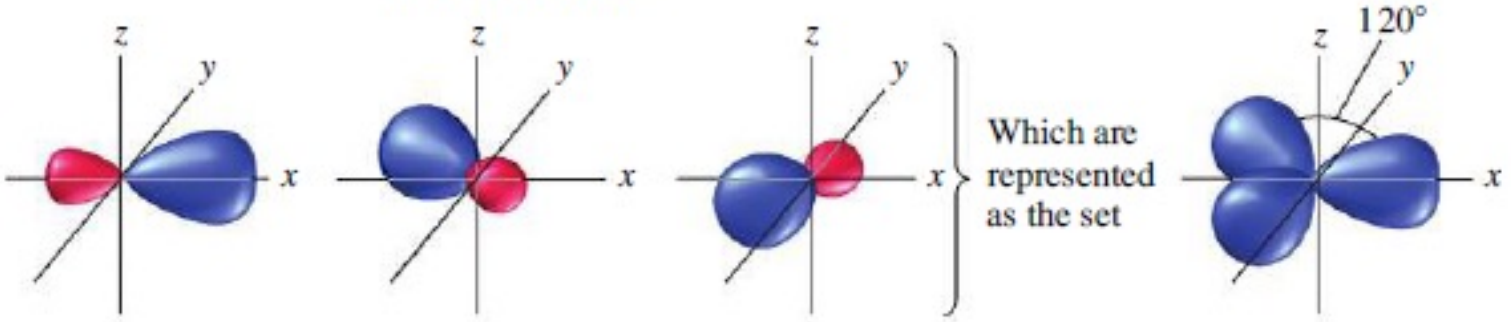
Sp^2 hybrid orbitals (example: B)



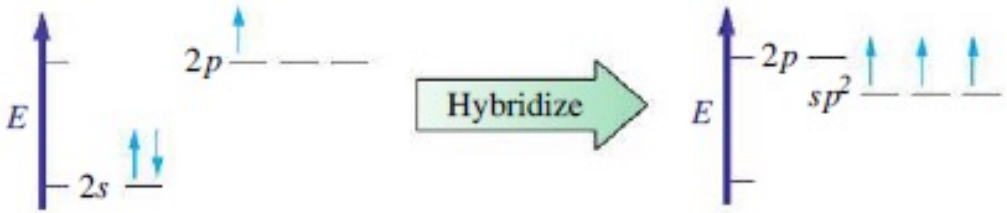
sp^2 hybridization scheme (example: BF_3)



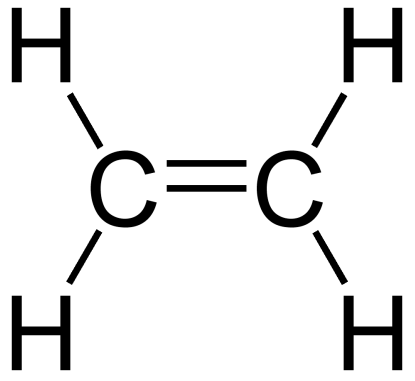


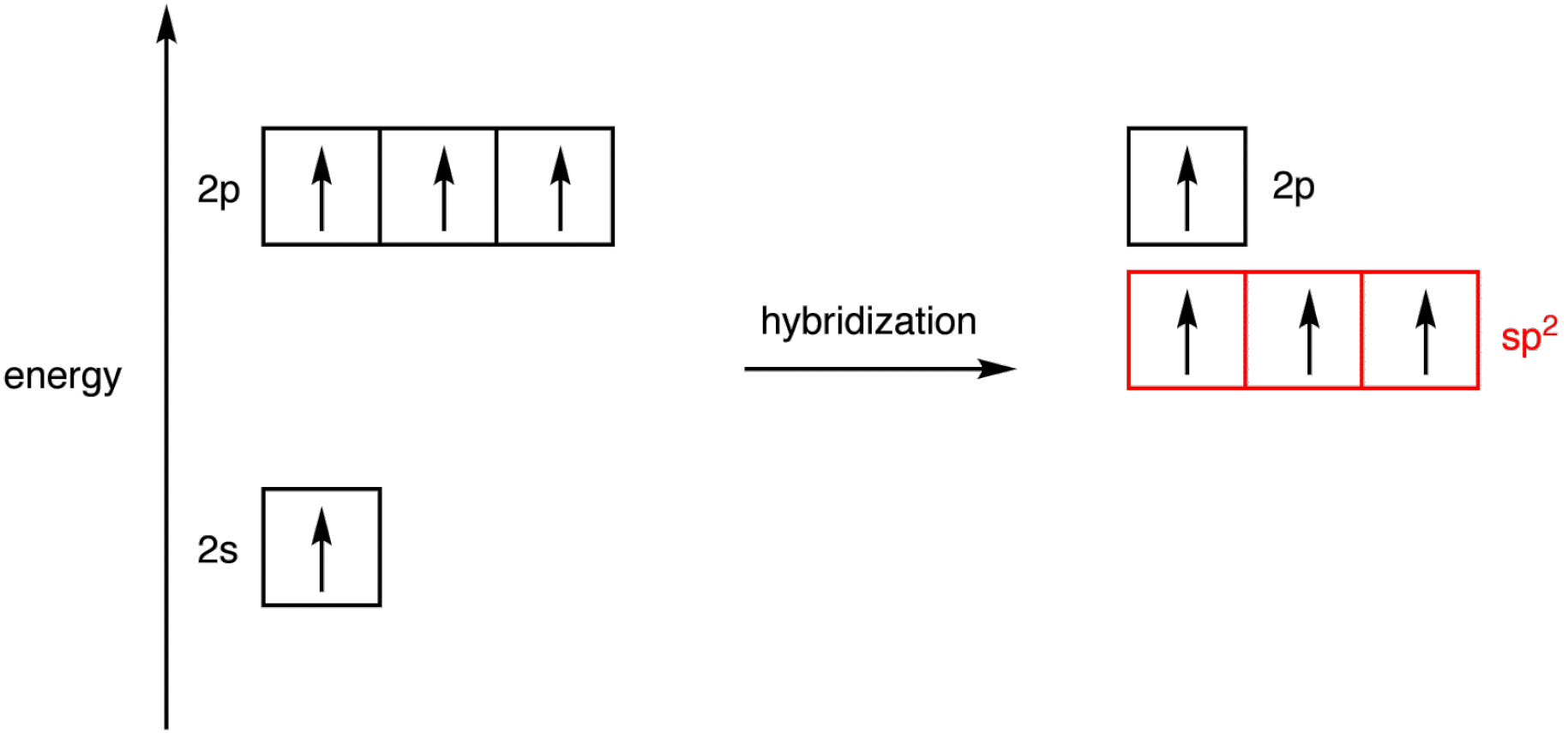


Identical bonds



Another example for sp^2 hybridization

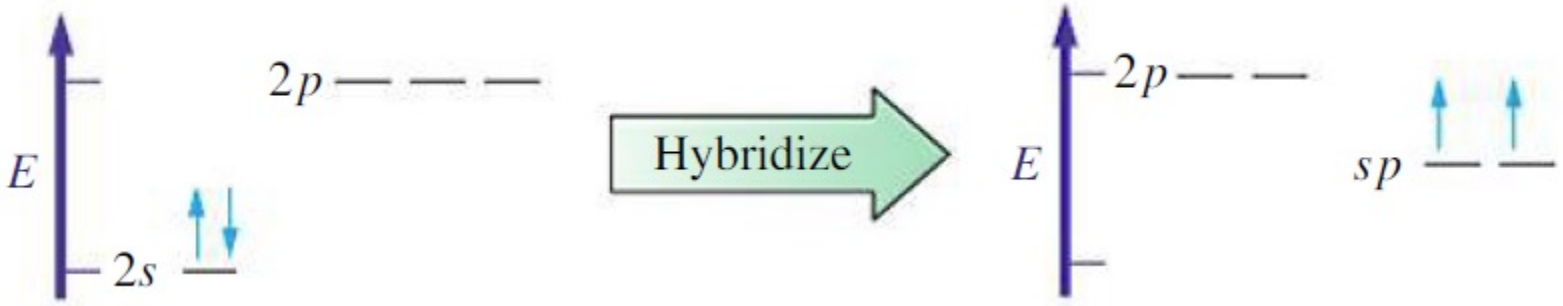




sp hybridization

Beryllium (Be) has four orbitals and only two valence electrons ($1s^2 2s^2$). Since it does not have half filled orbitals it is unlikely to form bonds.

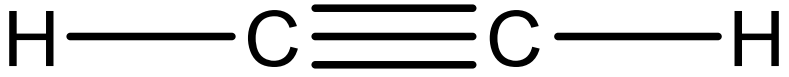
In certain gaseous beryllium compounds the 2s and 2p orbitals of Be hybridize into two sp hybrid orbitals and two p orbitals remain vacant.

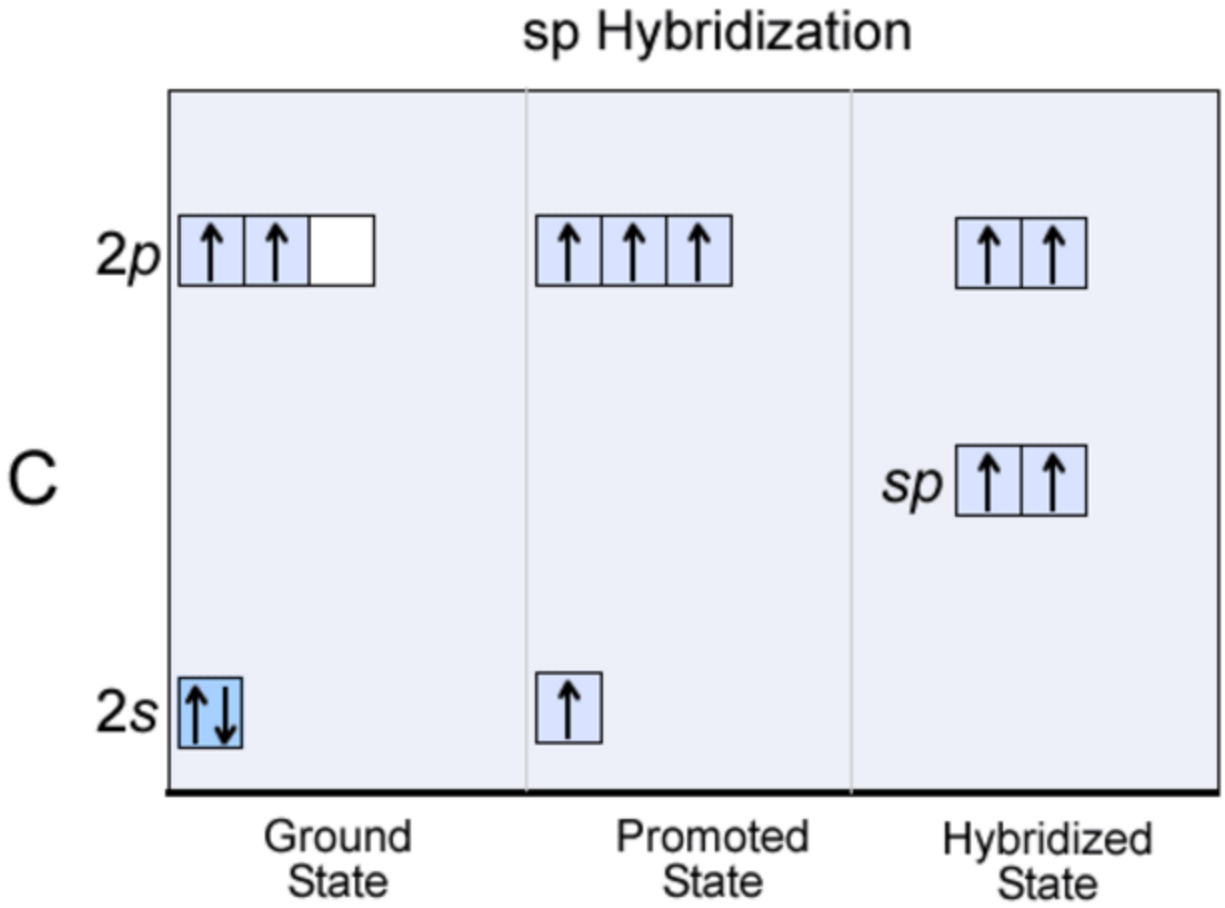


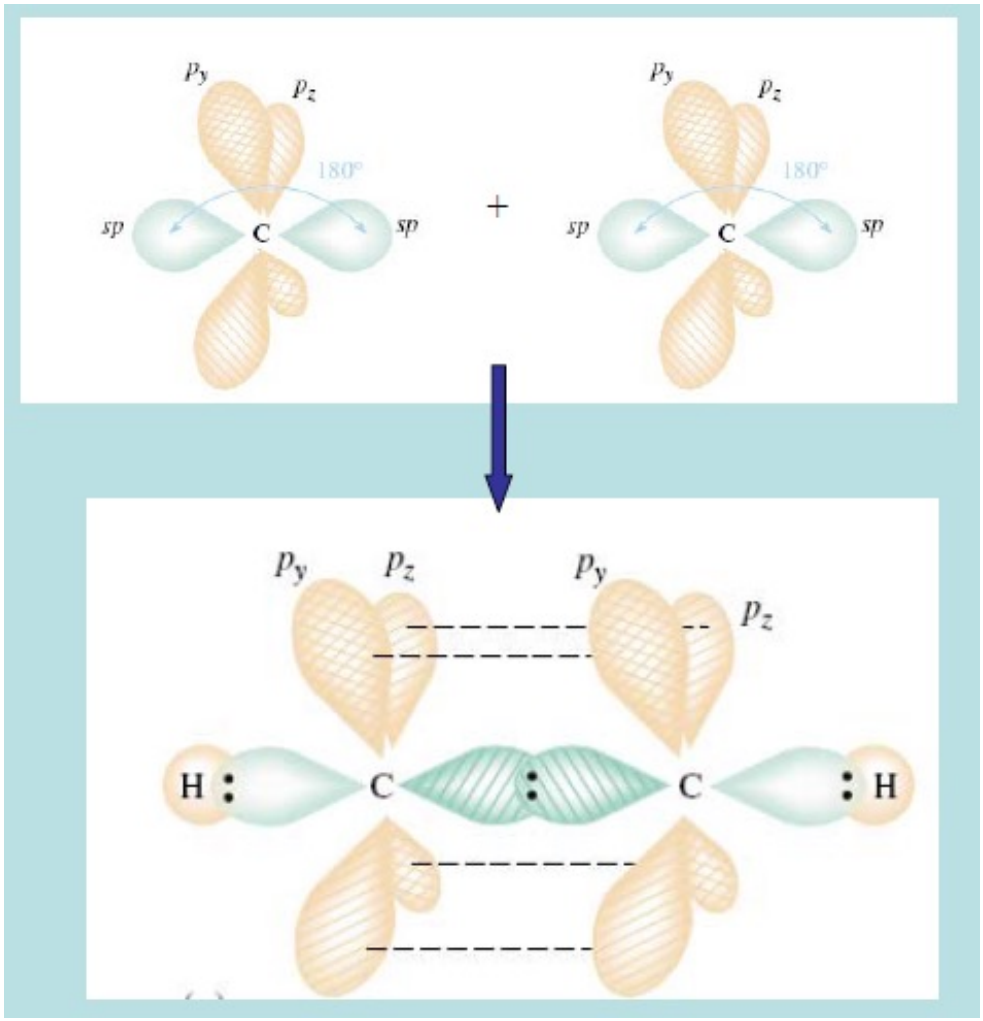
This hybridization scheme corresponds to a linear electron-group geometry and a 180° bond angle, such as in $BeCl_{2(g)}$, $BeBr_{2(g)}$, $Be(C_2H_5)_{2(g)}$ all with $1s^2 2s^1 2p_x^1$ config.

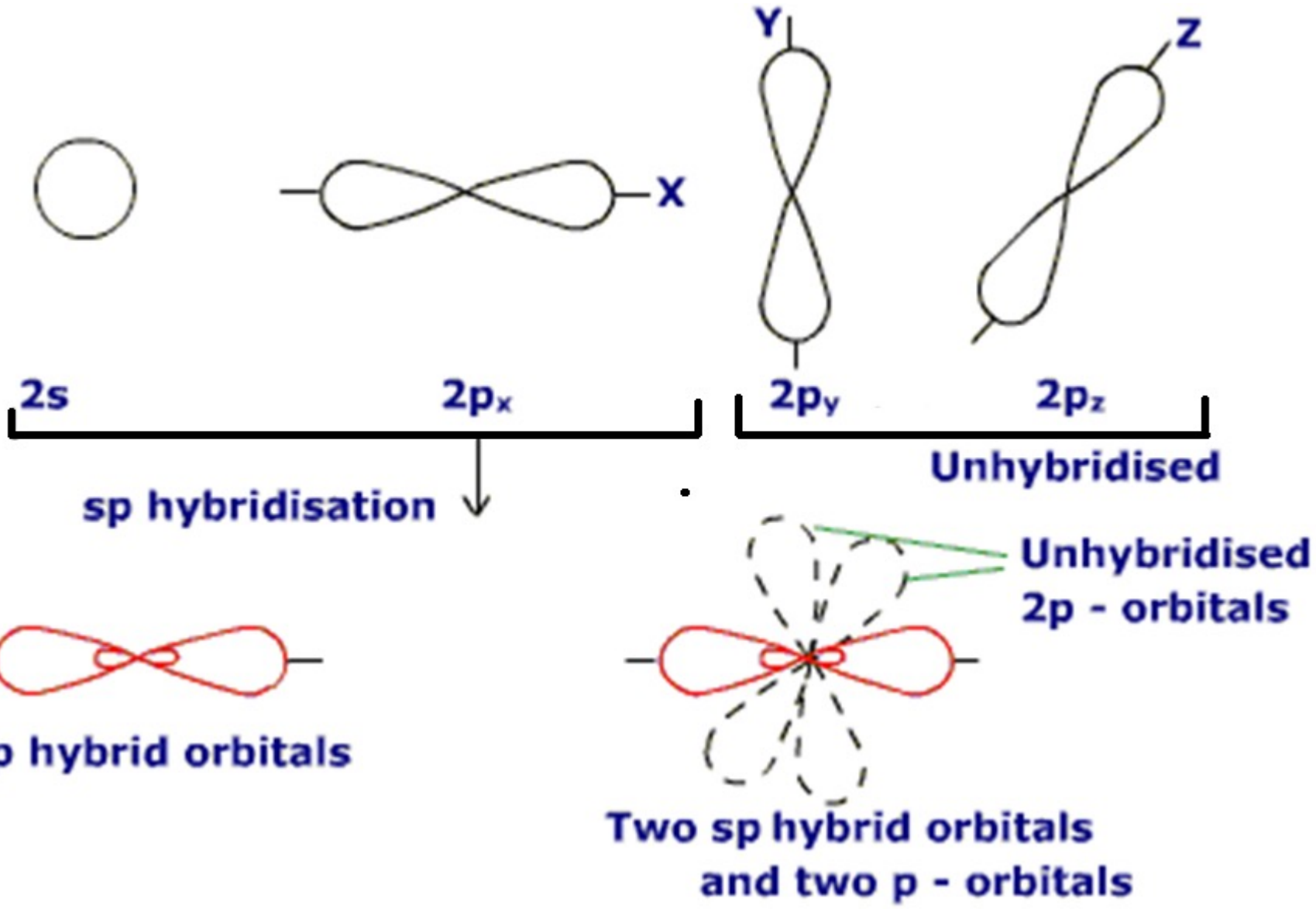
Another example of sp hybridization

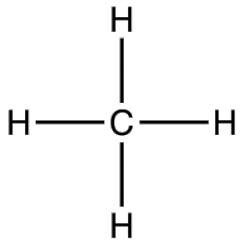
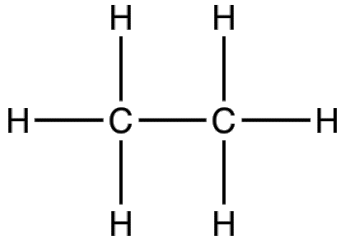
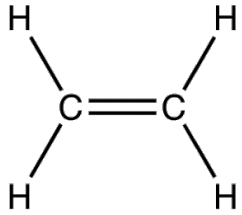
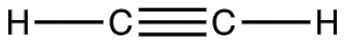
acetylene



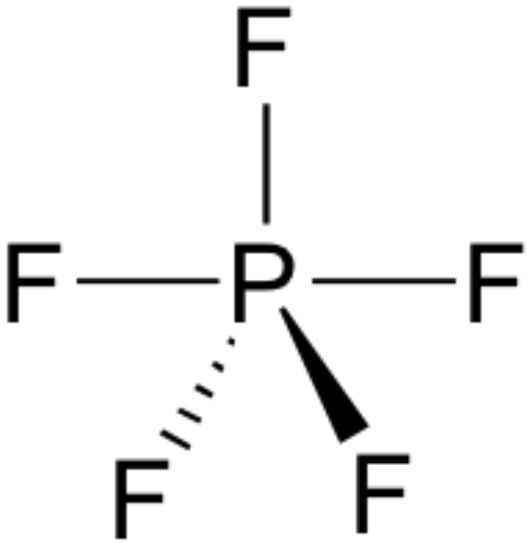


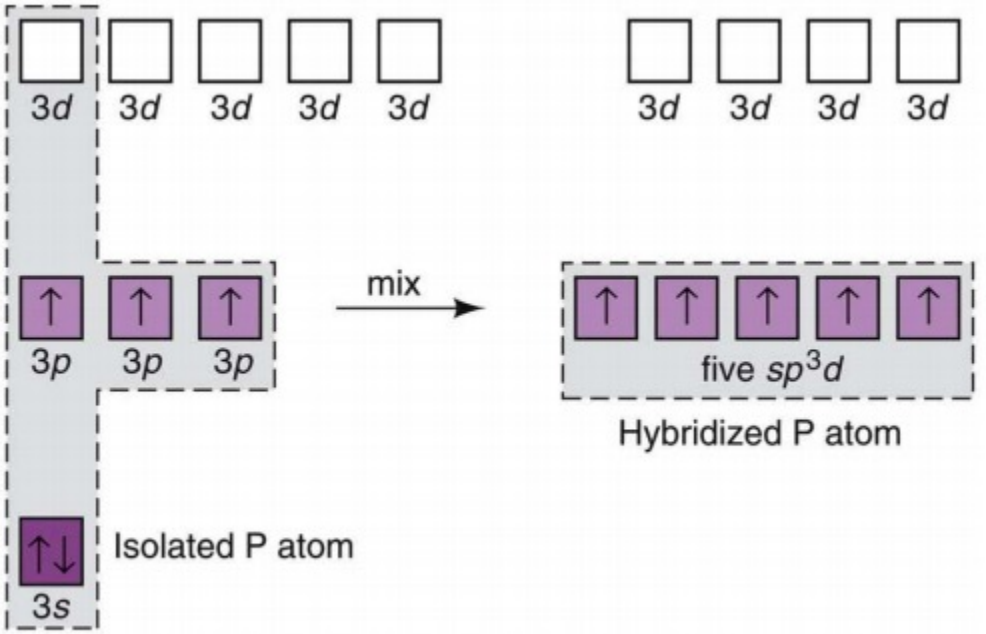

p_y
p_z
180°
sp
C
sp
+
p_y
p_z
180°
sp
C
sp
p_y
p_z
p_y
p_z
H
C
C
H


Y
Z
X
2s
2p$_x$
2p$_y$
2p$_z$
Unhybridised
sp hybridisation
Unhybridised
2p - orbitals
sp hybrid orbitals
Two sp hybrid orbitals
and two p - orbitals

compound	hybridization	geometry
	sp^3	tetrahedral
	sp^3	tetrahedral
	sp^2	planar trigonal
	sp	linear

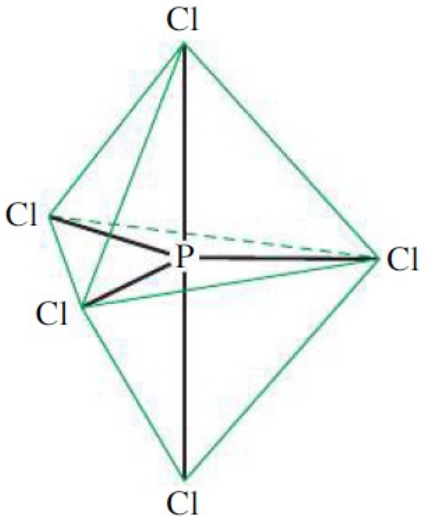
sp^3d hybridization







sp^3d orbitals



Trigonal-bipyramidal structure

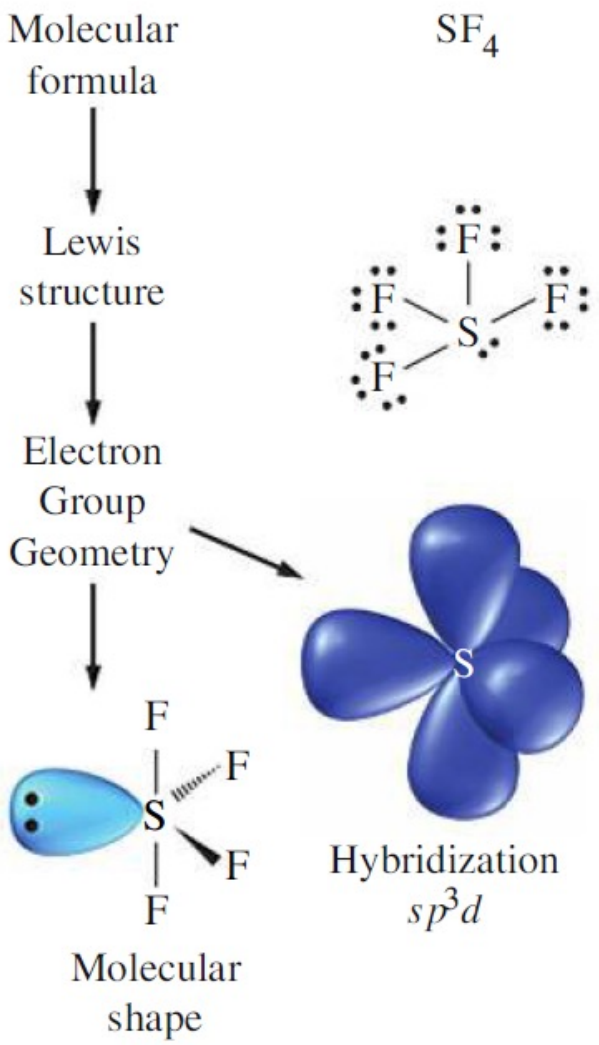


Procedure for suggesting hybrid orbitals and molecular geometry based on the VSEPR theory

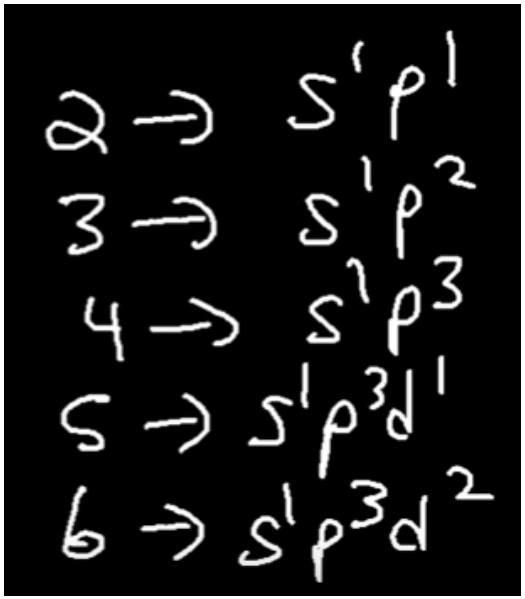
1. Write a plausible Lewis structure for the species of interest

2. Use VSEPR theory to predict the probable electron group geometry of the central atom

3. Select the hybridization scheme corresponding to the electron group geometry



Electron groups and corresponding hybridization scheme

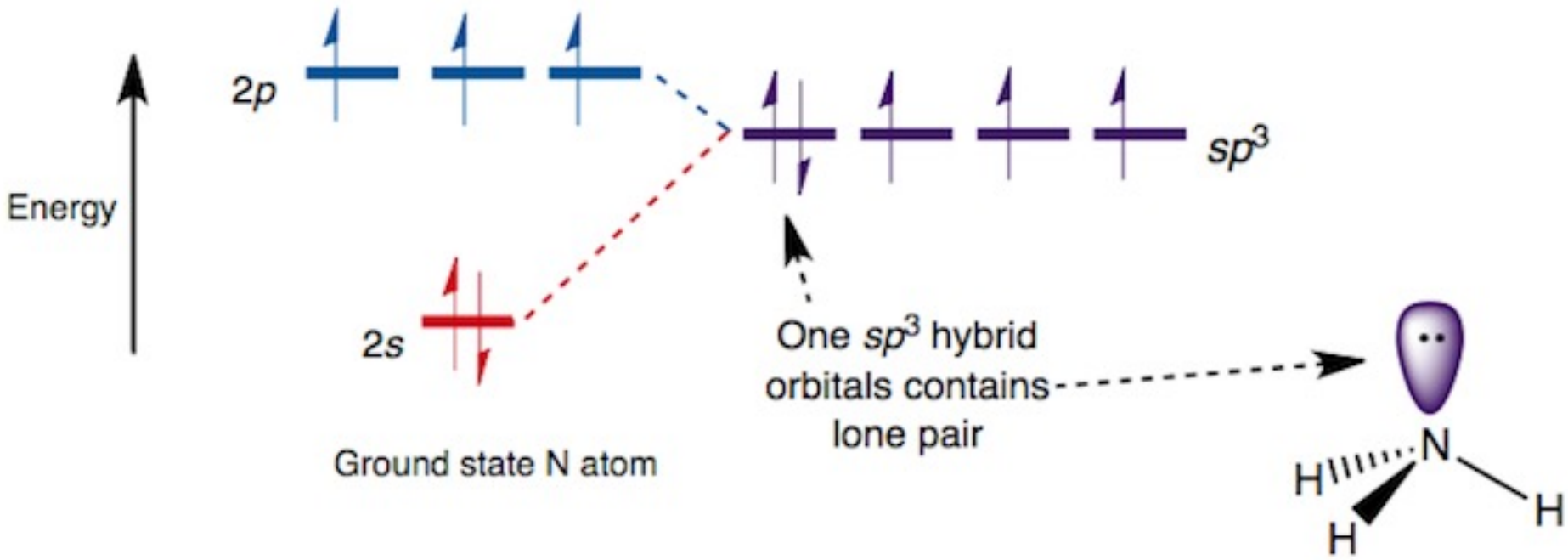


Hybrid orbitals and their geometric orientations

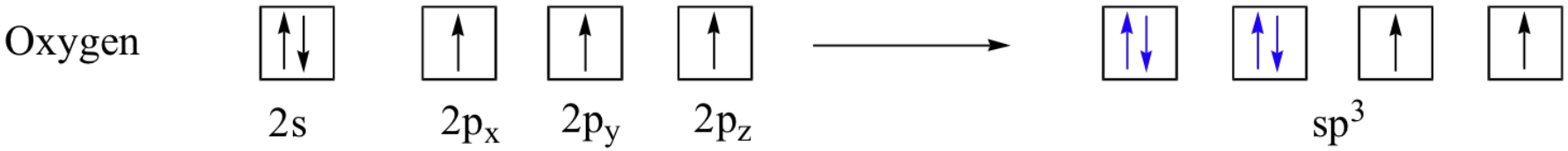
Hybrid Orbitals	Geometric Orientation	Example
sp	Linear	$BeCl_2$
sp^2	Trigonal-planar	BF_3
sp^3	Tetrahedral	CH_4
sp^3d	Trigonal-bipyramidal	PCl_5
sp^3d^2	Octahedral	SF_6

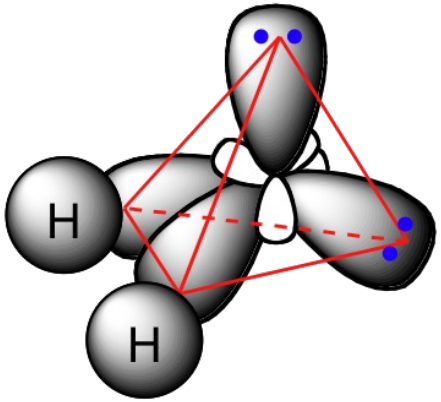
Particular examples

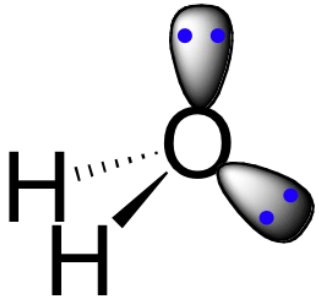
NH_3

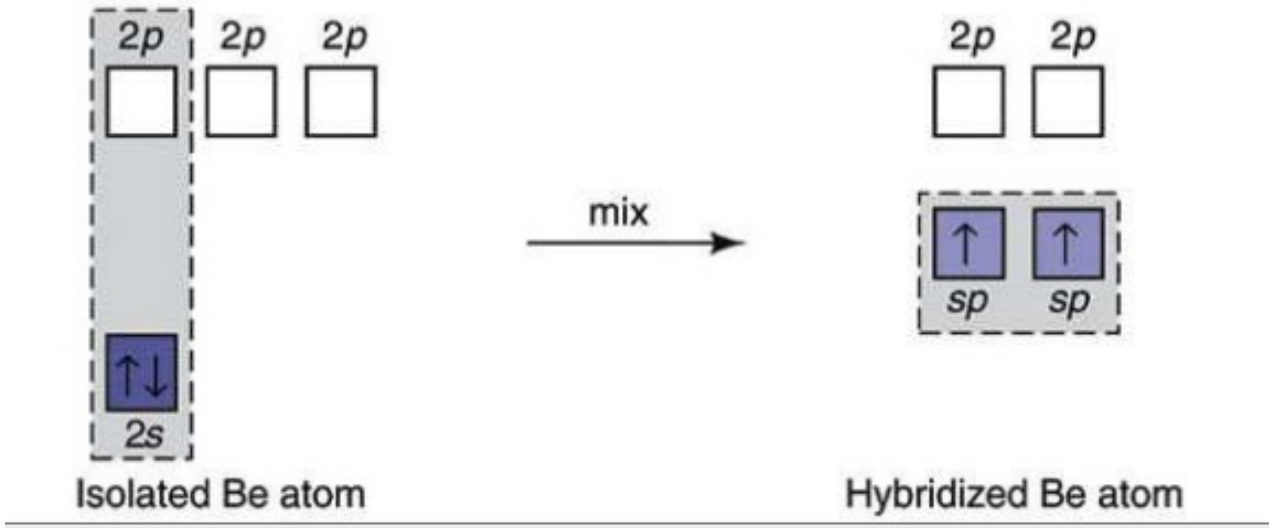


H_2O

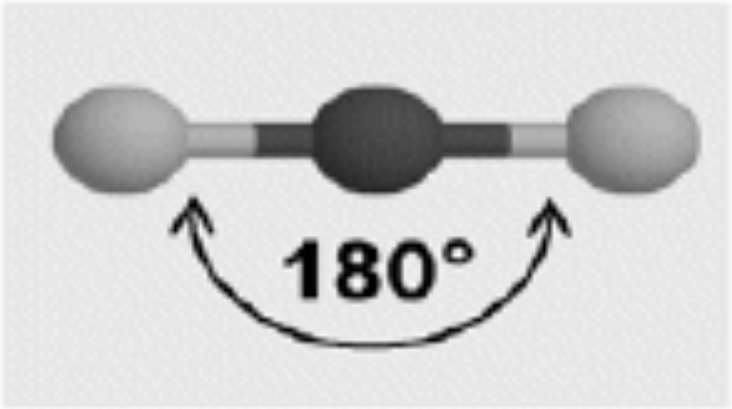


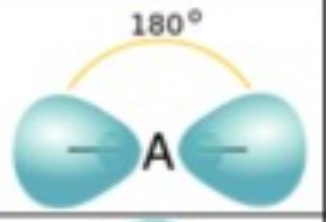
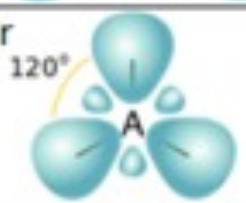




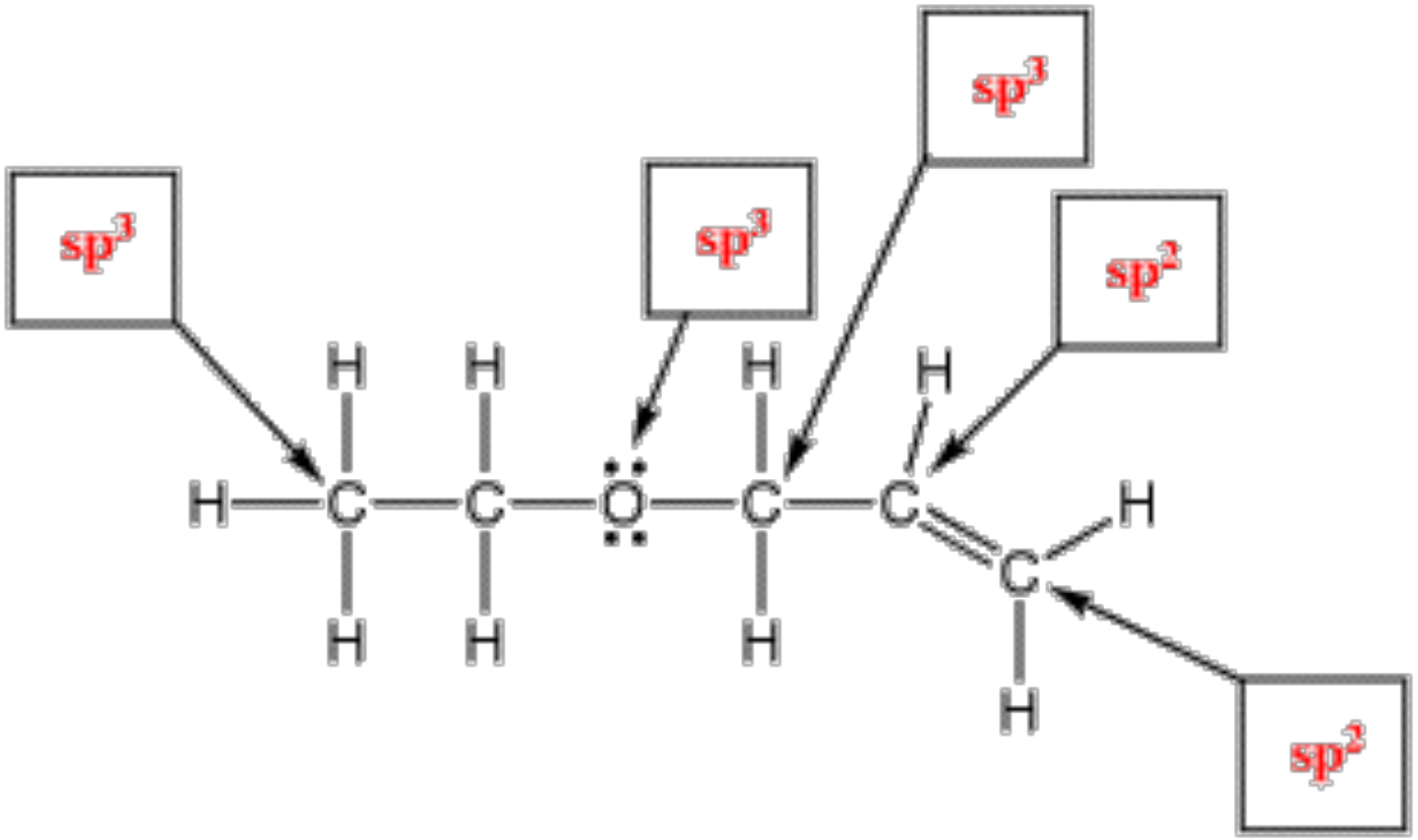

2p
2p
2p
2s
mix
2p
2p
sp
sp
Isolated Be atom
Hybridized Be atom

Cl–Be–Cl

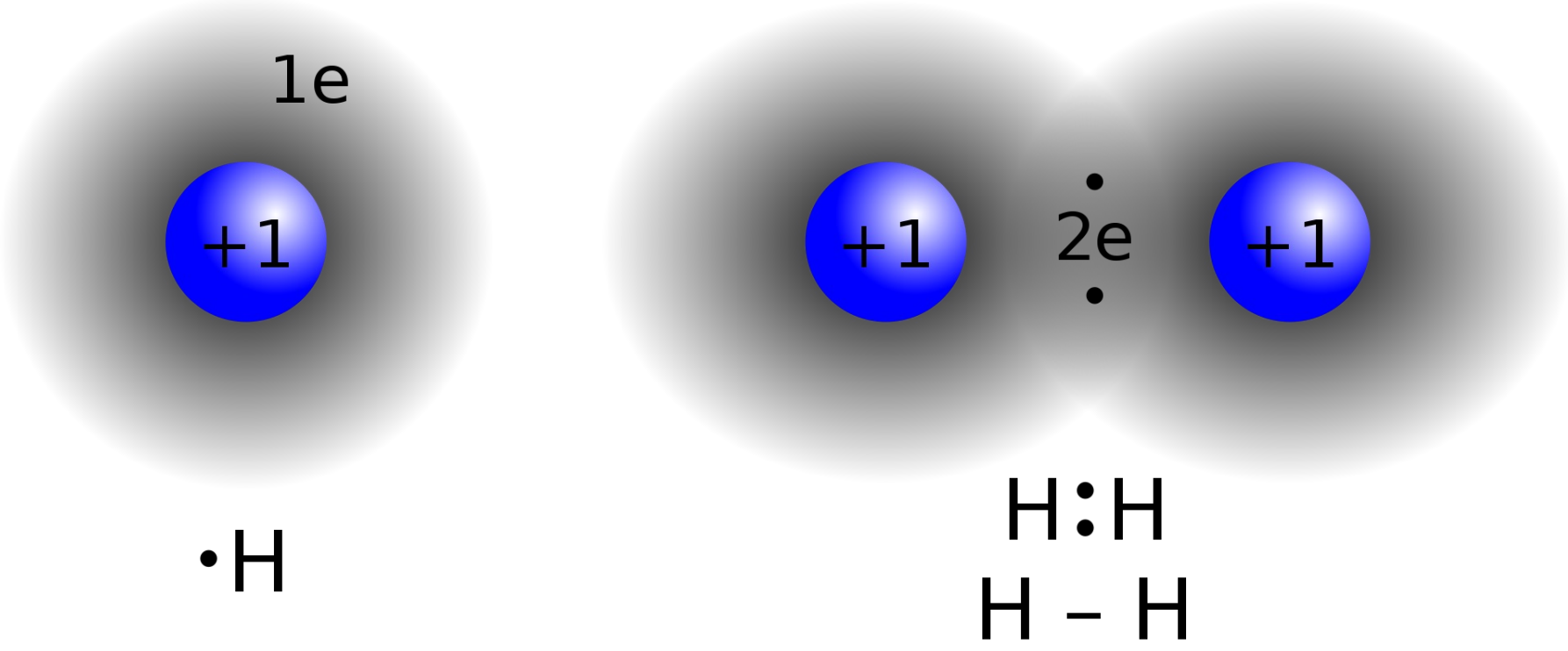

180°

Hybrid	Groups of Electrons	Angles	Shape
Sp	2	180 degrees	Linear 
Sp^2	3	120 degrees	Trigonal planar 
Sp^3	4	109.5 degrees	tetrahedral 

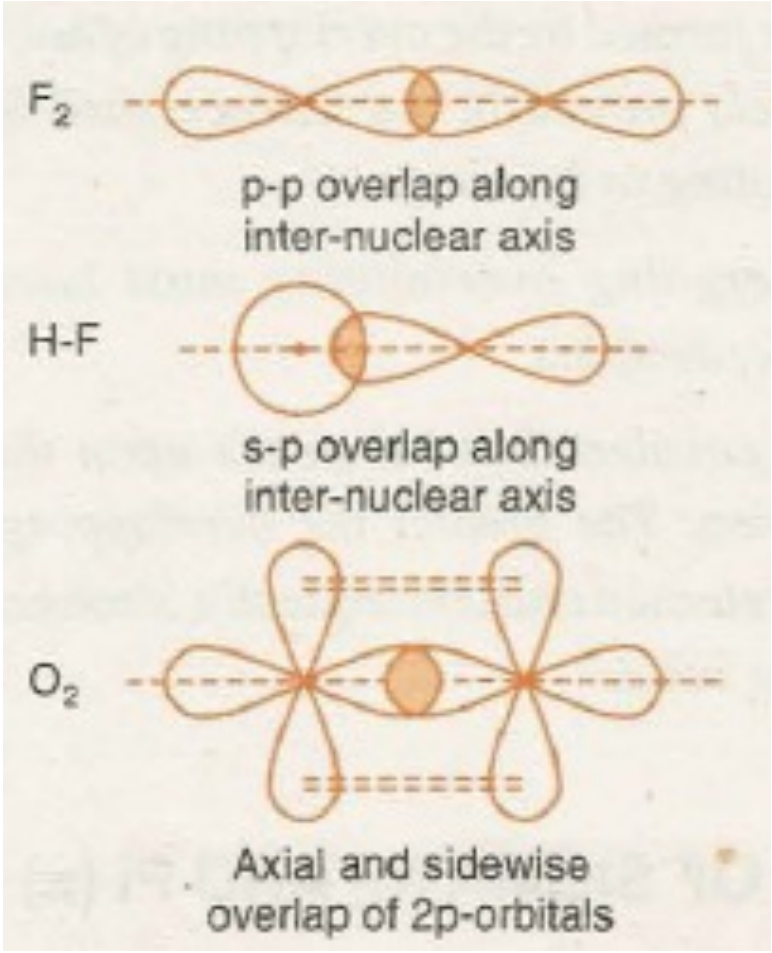
$C_5H_{10}O$



Further aspects of bonding

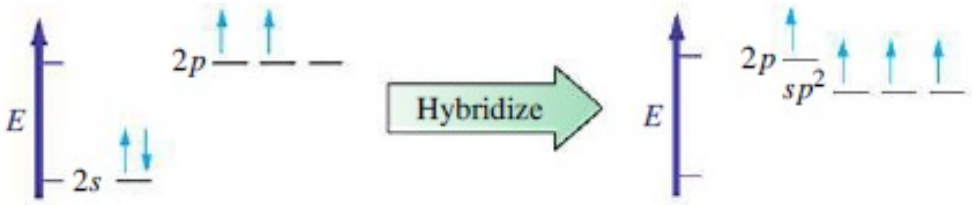


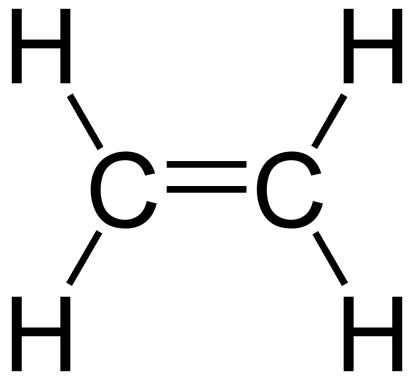
Orbital overlap



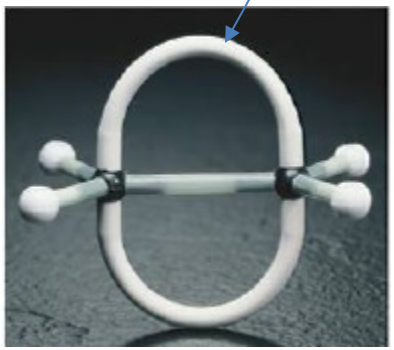
Multiple covalent bonds

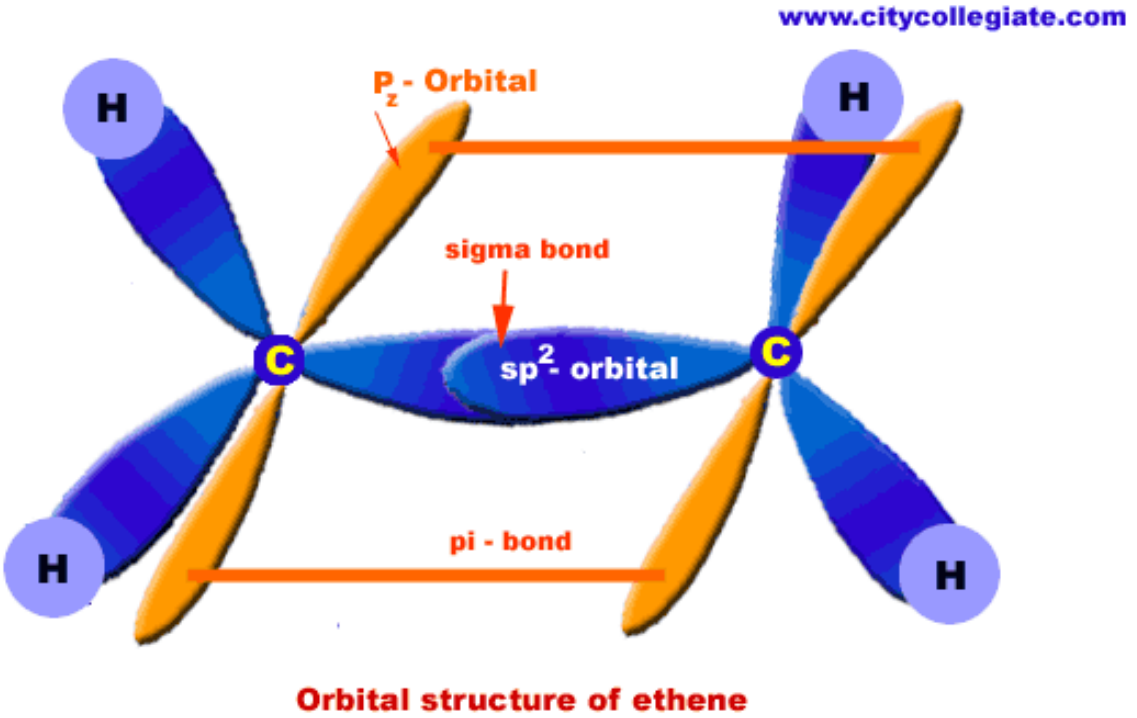
Ethane (double bond)





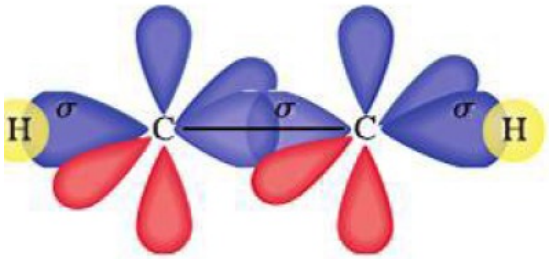
Overlapping of the un-hybridized p orbitals (termed “pi bond”)





The double bond is made up from one π bond and one σ bond

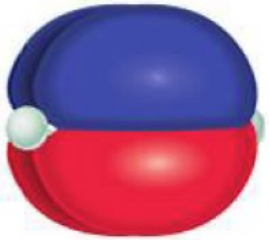
In a **triple bond** one of the carbon to carbon bonds is a σ bond and the other two are π bonds



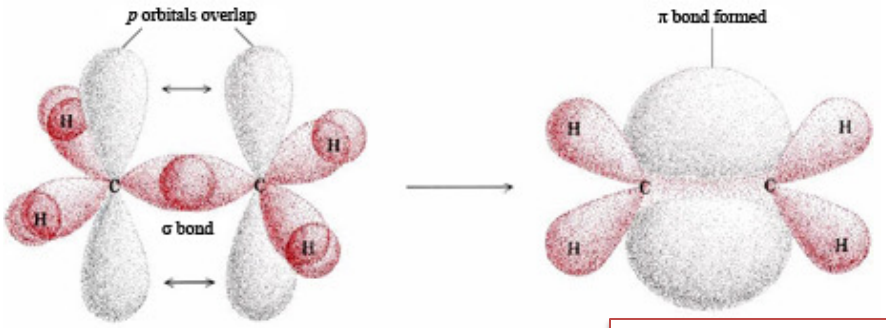
Formation of σ bonds

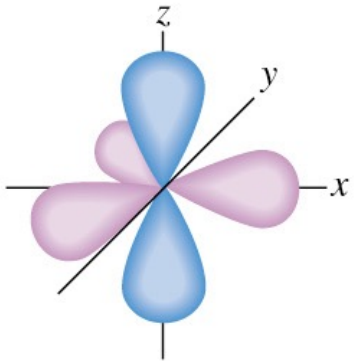


Formation of π bonds

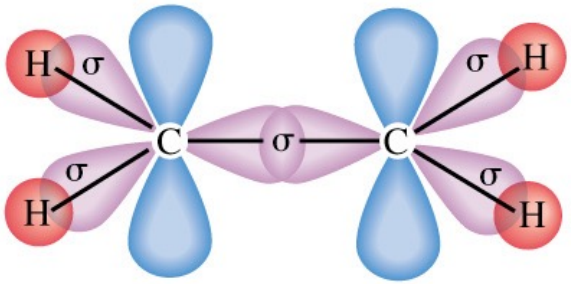

Computed π orbitals of acetylene

σ and π bonds

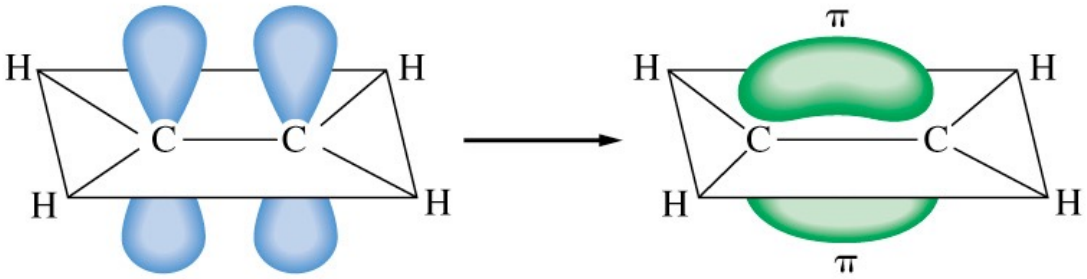




The set of orbitals $sp^2 + p$

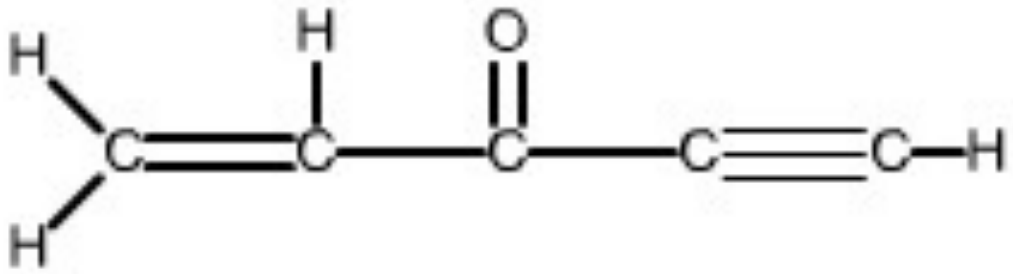


Sigma (σ) bonds



Overlap of *p* orbitals leading to pi (π) bond

How many σ and π bonds are in this molecule?



9 Sigma bonds

4 Pi bonds

1 single bond = 1 sigma bond

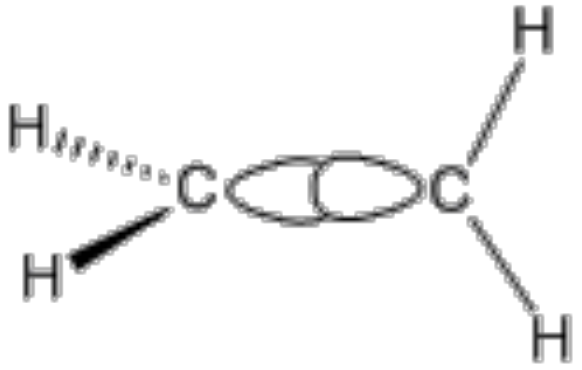
1 double bond = 1 sigma bond + 1 pi bond

1 triple bond = 1 sigma bond + 2 pi bonds

A single bond rotates freely

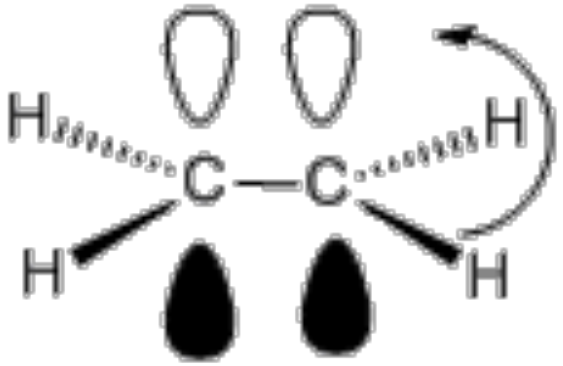


rotation about sigma bond

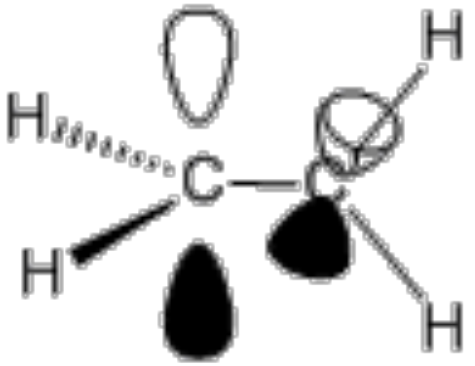


no change in orbital overlap

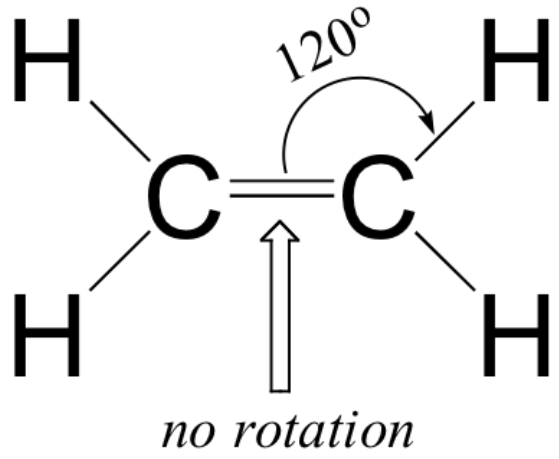
Double and Triple bonds have restricted rotation

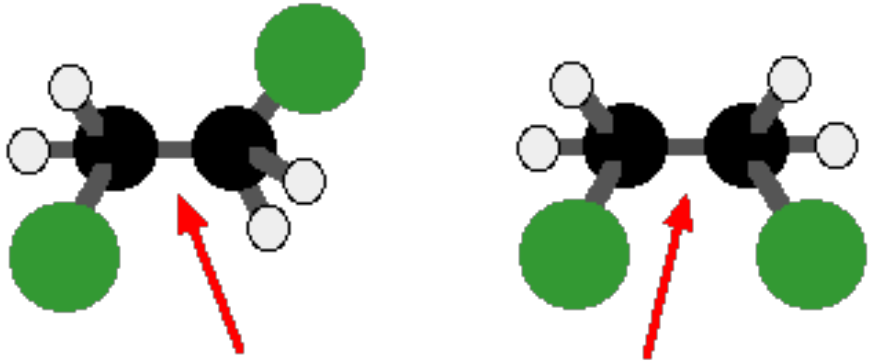


rotation about sigma bond



loss of orbital overlap
loss of bonding

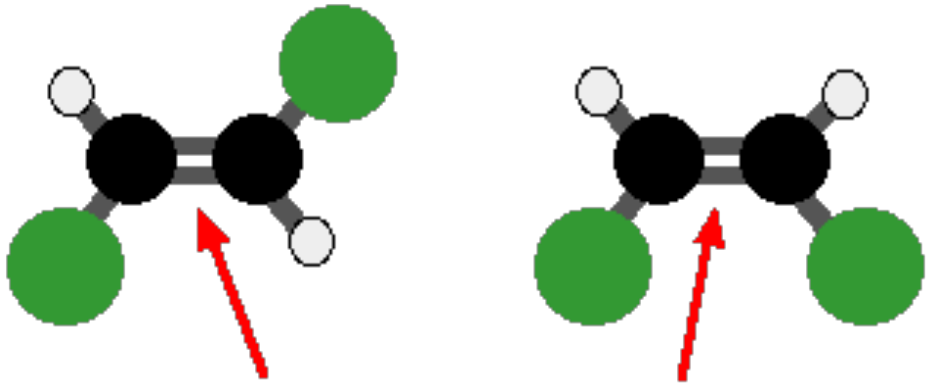




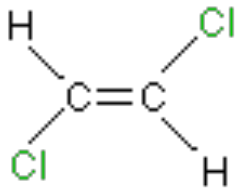
free rotation about this single bond

Exactly the same molecule. For example:

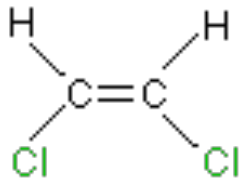




no rotation about this double bond



trans-1,2-dichloroethene



cis-1,2-dichloroethene

These are isomers (not the same molecule)

cis isomer means “on the same side of the double bond”